arXiv:2611.07362v1 [math.OC] 5 Nov 2026

Continuous-time q-Learning for Finite-Horizon Portfolio Choice with Optimal Stopping in Jump-Diffusion Markets

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We develop a structured continuous-time q-learning framework for finite-horizon investment and optimal stopping under jump-diffusion dynamics with unknown parameters. A penalty approximation transforms the Hamilton–Jacobi–Bellman variational inequality into an HJB equation with a binary control for the stopping penalty. An auxiliary state incorporates the resulting control-dependent discount factor into a Markovian formulation. We randomize controls on the mixed action space and introduce Shannon entropy regularization, establishing policy improvement and martingale characterizations of the continuous-time q-function. These results motivate a structured q-learning algorithm that constructs the q-function from the penalty HJB equation and jointly learns the value function and unknown generator parameters from sample paths. We apply the method to portfolio choice with optimal stopping in a simulation study. We further conduct a two-asset empirical study, evaluating learned strategies on bootstrap paths from held-out test returns versus an MLE-finite difference benchmark.

Keywords: Optimal investment and stopping; Continuous-time reinforcement learning; q-learning; Mixed control; Jump-diffusion

JEL Classification: G11; C61; C63

1. Introduction

Mixed control problems combine regular control with optimal stopping. The decision maker continuously adjusts the system through a control process and chooses a stopping time at which to terminate the activity and receive a payoff. Such problems arise in applications where ongoing decisions affect the state and the value of continuing, including portfolio management with an exit option. In this example, an investor adjusts the fraction of wealth invested in a risky asset while choosing when to leave the market. We study finite-horizon mixed control problems in which the controlled state may experience jumps.

When the market model and its parameters are known, the dynamic programming principle (DPP) characterizes the combined control and stopping problem via a Hamilton–Jacobi–Bellman (HJB) variational inequality (VI) with a nonlocal term that captures jump effects. Fleming and Soner (2006) and Øksendal and Sulem (2007) establish the theoretical foundation for stochastic control, jump-diffusion dynamics, and optimal stopping, which underpins such HJB variational inequalities. Building on this analytical framework, finite-difference and penalty methods compute value functions and corresponding policies for problems with low-dimensional state spaces. For

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example, Forsyth and Vetzal (2002) examine a numerical penalty approach for American option valuation, where the exercise decision gives rise to a free boundary. These methods, however, generally require full knowledge of model parameters. When parameters remain unknown, researchers commonly estimate parameters from observed data first and then solve the resulting VI. This framework separates parameter estimation and control design into two successive stages.

Continuous-time RL offers an alternative approach to solving stochastic control problems in unknown environments using sample paths. Wang *et al.* (2020) formulated an exploratory control problem through control randomization and entropy regularization. For an insurance portfolio management problem under diffusion dynamics, Cheng *et al.* (2026) combined entropy-regularized relaxed control with Markov chain approximation and stochastic approximation to develop a hybrid deep reinforcement learning (DRL) method for reinsurance and dividend decisions. Jia and Zhou (2022a) characterized the value function of a given policy through martingale conditions, thereby expressing policy evaluation as a martingale learning problem based on observed sample paths. Building on this approach, Jia and Zhou (2022b) developed continuous-time policy gradient and actor–critic algorithms. To address the degeneration of the conventional Q-function to the value function in the continuous-time limit, Jia and Zhou (2023) defined a continuous-time q-function through a first-order expansion under a short-time action perturbation and established a joint martingale characterization of the value function and q-function.

Recent work has extended these ideas to jump-diffusion control and optimal stopping. Gao *et al.* (2026) extended RL methods based on controlled generators and martingale conditions to jump-diffusion models, while Bo *et al.* (2026) studied continuous-time q-learning for such models under Tsallis entropy regularization. For optimal stopping, Dong (2024) developed a continuous-time RL algorithm by randomizing stopping decisions. Starting from a penalty approximation of the associated VI, Dai *et al.* (2026) represented the penalty term using a binary control and combined Bernoulli randomization with martingale-based policy evaluation to learn stopping rules. These studies address learning for continuous control with jumps and for optimal stopping of diffusion processes, respectively. They do not, however, provide a unified framework incorporating continuous-valued investment control, stopping decisions and jump risk. When market parameters are unknown, we further seek a framework that learns these parameters jointly with the value function and derives the associated investment and stopping rules.

Motivated by these developments, we follow the penalty approach in Dai *et al.* (2026) to approximate the HJBVI by an HJB equation whose binary control represents the stopping penalty. Together with the original regular control, this yields a mixed control problem with a control-dependent discount factor, which we incorporate into an augmented Markovian state. Building on the exploratory control and continuous-time q-learning framework of Jia and Zhou (2023), we then randomize the controls on the mixed action space and introduce Shannon entropy regularization. We establish policy improvement and martingale characterizations of the continuous-time q-function, accounting for both the nonlocal jump generator and the stopping penalty. These results underpin a structured q-learning algorithm that constructs the q-function from the penalty HJB equation and jointly updates the value-function parameters and unknown generator parameters from sample paths using martingale orthogonality conditions (Jia and Zhou 2022a). We subsequently apply this framework to portfolio choice with optimal stopping in a jump-diffusion market. A simulation study examines the learned value function, investment policy, stopping boundary, and market parameters against finite-difference benchmarks. We further conduct an empirical study using the Huatai-PineBridge CSI 300 ETF (510300) and the ARK Innovation ETF (ARKK), evaluating the learned investment and stopping rules on bootstrap paths drawn from held-out test-period returns and comparing them with a benchmark combining maximum likelihood estimation and finite-difference methods.

The contributions are threefold. First, we cast the finite-horizon optimal investment and stopping problem under jump-diffusion dynamics into a penalized mixed-control formulation, where an auxiliary state variable is introduced to encode the control-dependent discounting factor. Second, for the Shannon-entropy-regularized counterpart, we develop policy improvement theorems and

on-policy as well as off-policy martingale representations for the value function and q-function. These characterizations explicitly incorporate the nonlocal jump generator and the penalty associated with the stopping constraint. Third, we extract a structurally parameterized q-function from the penalty HJB equation, which enables joint learning of the value function and unknown generator parameters directly from trajectory samples. We validate the numerical performance of our approach via simulation benchmarks and a two-asset bootstrap experiment, focusing on portfolio optimization within jump-diffusion markets.

The remainder of the paper is organized as follows. Section 2 develops the penalty approximation and mixed-control representation. Section 3 formulates the Shannon-entropy-regularized exploratory problem and establishes policy improvement. Section 4 defines the continuous-time q-function and proves its martingale characterizations. Section 5 develops the structured q-learning algorithm and joint parameter updates. Section 6 presents the financial application and numerical experiments, and Section 7 concludes. Proofs and the finite-difference method are provided in the appendices.

2. Formulation of the Mixed Control Problem

This section formulates a finite-horizon control problem with optimal stopping in a jump-diffusion model. We first derive the associated HJBVI, then approximate its stopping obstacle by a penalty term. The penalized equation is represented as a mixed control problem, which we subsequently express in an augmented Markovian form.

2.1. *Model setup and assumptions*

Consider a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P}; (\mathcal{F}_t)_{t\geq 0})$ satisfying the usual conditions. Let $\boldsymbol{W} = (\boldsymbol{W}_t)_{t\geq 0}$ be a d-dimensional standard Brownian motion and $N(\mathrm{d}t, \mathrm{d}z)$ an independent Poisson random measure on $[0,T]\times\mathbb{R}$ with intensity measure $\mathrm{d}t\, m(\mathrm{d}z)$, where m is a finite measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Fix a finite horizon $[0,T]$ and a discount rate $\rho > 0$, and let $A \subset \mathbb{R}^k$ be the investment action space. For each $s \in [0,T]$, denote by $\mathcal{A}_s$ the set of admissible A-valued, $(\mathcal{F}_t)_{t\geq 0}$-predictable investment control processes $\boldsymbol{\alpha} = (\boldsymbol{\alpha}_t)_{t\in[s,T]}$, so that investment decisions use only information available before time t. Denote by $\mathcal{T}_{s,T}$ the set of $(\mathcal{F}_t)_{t\geq 0}$-stopping times taking values in $[s,T]$. The finite-horizon optimal investment and stopping problem is

$$v(s,\boldsymbol{x}) = \sup_{\boldsymbol{\alpha}\in\mathcal{A}_s,\, \tau\in\mathcal{T}_{s,T}} \mathbb{E}_{s,\boldsymbol{x}}\left[\mathrm{e}^{-\rho(\tau-s)} g(\tau, \boldsymbol{X}^{\alpha}_{\tau})\right], \tag{1}$$

where $g(t,\boldsymbol{x})$ is the stopping payoff and the wealth process $\boldsymbol{X}^{\alpha}$ evolves according to

$$\mathrm{d}\boldsymbol{X}^{\alpha}_t = \boldsymbol{b}(t, \boldsymbol{X}^{\alpha}_t, \boldsymbol{\alpha}_t)\,\mathrm{d}t + \boldsymbol{\sigma}(t, \boldsymbol{X}^{\alpha}_t, \boldsymbol{\alpha}_t)\,\mathrm{d}\boldsymbol{W}_t + \int_{\mathbb{R}} \boldsymbol{\gamma}(t, \boldsymbol{X}^{\alpha}_{t-}, \boldsymbol{\alpha}_t, z)\, N(\mathrm{d}t, \mathrm{d}z), \quad \boldsymbol{X}^{\alpha}_s = \boldsymbol{x}. \tag{2}$$

Assumption 2.1 The control set, model coefficients, and stopping payoff satisfy:

(i) The control set $A \subset \mathbb{R}^k$ is nonempty and compact, with positive Lebesgue measure. The drift $\boldsymbol{b} : [0,T]\times\mathbb{R}^n \times A \to \mathbb{R}^n$, diffusion coefficient $\boldsymbol{\sigma} : [0,T]\times\mathbb{R}^n \times A \to \mathbb{R}^{n\times d}$, and jump amplitude $\boldsymbol{\gamma} : [0,T]\times\mathbb{R}^n \times A \times \mathbb{R} \to \mathbb{R}^n$ are measurable and continuous in $(t,\boldsymbol{x},\boldsymbol{\alpha})$. The stopping payoff $g : [0,T]\times\mathbb{R}^n \to \mathbb{R}$ is continuous.

(ii) There exist a fixed $p \geq 2$ and a constant $C > 0$ such that, for every $t \in [0,T]$, $\boldsymbol{x}, \boldsymbol{x}' \in \mathbb{R}^n$,

and $\boldsymbol{\alpha} \in A$,

$$
\begin{aligned}
&|g(t,\boldsymbol{x})| \le C(1+|\boldsymbol{x}|^p),\\
&|\boldsymbol{b}(t,\boldsymbol{x},\boldsymbol{\alpha}) - \boldsymbol{b}(t,\boldsymbol{x}',\boldsymbol{\alpha})| + \|\boldsymbol{\sigma}(t,\boldsymbol{x},\boldsymbol{\alpha}) - \boldsymbol{\sigma}(t,\boldsymbol{x}',\boldsymbol{\alpha})\| \le C|\boldsymbol{x}-\boldsymbol{x}'|,\\
&\left(\int_{\mathbb{R}} |\boldsymbol{\gamma}(t,\boldsymbol{x},\boldsymbol{\alpha},z) - \boldsymbol{\gamma}(t,\boldsymbol{x}',\boldsymbol{\alpha},z)|^2\, m(\mathrm{d}z)\right)^{1/2} \le C|\boldsymbol{x}-\boldsymbol{x}'|,\\
&|\boldsymbol{b}(t,\boldsymbol{x},\boldsymbol{\alpha})| + \|\boldsymbol{\sigma}(t,\boldsymbol{x},\boldsymbol{\alpha})\| \le C(1+|\boldsymbol{x}|),\\
&\left(\int_{\mathbb{R}} |\boldsymbol{\gamma}(t,\boldsymbol{x},\boldsymbol{\alpha},z)|^p\, m(\mathrm{d}z)\right)^{1/p} \le C(1+|\boldsymbol{x}|).
\end{aligned}
$$

Under Assumption 2.1, (2) admits a unique strong solution for every $\boldsymbol{\alpha} \in \mathcal{A}_s$. Moreover, $\mathbb{E}_{s,\boldsymbol{x}}[\sup_{r\in[s,T]} |\boldsymbol{X}_r^{\alpha}|^p] \le C_{p,T}(1+|\boldsymbol{x}|^p)$ for some constant $C_{p,T} > 0$ independent of $\boldsymbol{\alpha}$. Together with the growth condition on g, this estimate ensures that the payoff in (1) is absolutely integrable for every admissible control–stopping pair and that the value function is finite.

By the DPP, the value function of (1) satisfies the HJBVI

$$
\begin{cases}
\max\left\{ v_t - \rho v + \sup\limits_{\boldsymbol{\alpha}\in A}\big(\mathcal{L}^{\alpha} v + \mathcal{J}^{\alpha} v\big),\ g - v \right\} = 0, \quad (t,\boldsymbol{x}) \in [0,T)\times\mathbb{R}^n,\\
v(T,\boldsymbol{x}) = g(T,\boldsymbol{x}),
\end{cases}
\tag{3}
$$

where the operator $\mathcal{L}^{\alpha}$ and the nonlocal jump operator $\mathcal{J}^{\alpha}$ are defined by

$$
\begin{aligned}
\mathcal{L}^{\alpha} v(t,\boldsymbol{x}) &= \boldsymbol{b}(t,\boldsymbol{x},\boldsymbol{\alpha})\cdot v_{\boldsymbol{x}}(t,\boldsymbol{x}) + \frac{1}{2}\operatorname{tr}\big[\boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathrm{T}}(t,\boldsymbol{x},\boldsymbol{\alpha}) v_{\boldsymbol{x}\boldsymbol{x}}(t,\boldsymbol{x})\big],\\
\mathcal{J}^{\alpha} v(t,\boldsymbol{x}) &= \int_{\mathbb{R}} \big[v\big(t,\boldsymbol{x} + \boldsymbol{\gamma}(t,\boldsymbol{x},\boldsymbol{\alpha},z)\big) - v(t,\boldsymbol{x})\big]\, m(\mathrm{d}z).
\end{aligned}
\tag{4}
$$

2.2. *Penalty approximation*

To handle the obstacle in (3), we approximate it by a penalty term with parameter $K > 0$. Let v^K denote the solution of the resulting penalized equation, which also underlies numerical methods for free-boundary problems (Forsyth and Vetzal 2002, Dai *et al.* 2007, 2026):

$$
\begin{cases}
v_t - \rho v + \sup\limits_{\boldsymbol{\alpha}\in A}\big(\mathcal{L}^{\alpha} v + \mathcal{J}^{\alpha} v\big) + K(g-v)^+ = 0, \quad (t,\boldsymbol{x}) \in [0,T)\times\mathbb{R}^n,\\
v(T,\boldsymbol{x}) = g(T,\boldsymbol{x}).
\end{cases}
\tag{5}
$$

Assumption 2.2 In addition to Assumption 2.1, impose the following conditions. For functions of $(t,\boldsymbol{x})$, write

$$
\|h\|_1 := \sup_{t,\boldsymbol{x}} |h(t,\boldsymbol{x})| + \sup_{\substack{t,\boldsymbol{x},\boldsymbol{x}'\\ \boldsymbol{x}\neq\boldsymbol{x}'}} \frac{|h(t,\boldsymbol{x}) - h(t,\boldsymbol{x}')|}{|\boldsymbol{x}-\boldsymbol{x}'|},
$$

where the suprema are taken over $t \in [0,T]$ and $\boldsymbol{x},\boldsymbol{x}' \in \mathbb{R}^n$. There exists $C > 0$ such that the following bounds hold for every $\boldsymbol{\alpha} \in A$ and $z \in \mathbb{R}$:

$$
\|\boldsymbol{b}(\cdot,\cdot,\boldsymbol{\alpha})\|_1 + \|\boldsymbol{\sigma}(\cdot,\cdot,\boldsymbol{\alpha})\|_1 + \|g\|_1 \le C, \quad \|\boldsymbol{\gamma}(\cdot,\cdot,\boldsymbol{\alpha},z)\|_1 \le C(1\wedge|z|).
$$

The following theorem establishes monotone convergence of the penalty approximation in the class of bounded continuous viscosity solutions under Assumptions 2.1 and 2.2.

THEOREM 2.3 *Suppose that Assumptions 2.1 and 2.2 hold. Let v and v^K be the bounded viscosity solutions of* (3) *and* (5)*, respectively. Then, as $K \to \infty$,*

$$v^K(t, \boldsymbol{x}) \uparrow v(t, \boldsymbol{x}), \qquad (t, \boldsymbol{x}) \in [0, T] \times \mathbb{R}^n.$$

The proof of Theorem 2.3 is given in Appendix A.1.The theorem shows that the original VI can be approximated by solving (5) for sufficiently large K. Throughout the following analysis, we fix $K > 0$ and write v instead of v^K for the value function of the penalized problem, restoring the superscript when considering the limit $K \to \infty$. Using the binary-control representation

$$K(g - v)^+ = \max_{\beta \in \{0,1\}} \{-K\beta v + K\beta g\},$$

we can equivalently express (5) as the HJB equation

$$0 = v_t - \rho v + \sup_{\boldsymbol{\alpha} \in A,\, \beta \in \{0,1\}} \big\{\mathcal{L}^\alpha v + \mathcal{J}^\alpha v - K\beta v + K\beta g\big\}, \tag{6}$$

where $\beta = 1$ and $\beta = 0$ correspond to stopping and continuing, respectively, in the original optimal stopping problem. For finite K, the choice $\beta = 1$ approximates immediate stopping through a termination mechanism with intensity K.

To describe the two controls jointly, define the mixed action space $U := A \times \{0, 1\}$. For $\boldsymbol{u} = (\boldsymbol{u}^\alpha, u^\beta) \in U$, $\boldsymbol{u}^\alpha \in A$ is the continuous-valued investment control and $u^\beta \in \{0, 1\}$ is the binary stopping control. Equation (6) then takes the form

$$0 = v_t - \rho v + \sup_{\boldsymbol{u} \in U} \big\{\mathcal{L}^{\boldsymbol{u}^\alpha} v + \mathcal{J}^{\boldsymbol{u}^\alpha} v - K u^\beta v + K u^\beta g\big\}, \tag{7}$$

where the state dynamics depend only on $\boldsymbol{u}^\alpha$, while u^β enters through the penalty term $Ku^\beta(g-v)$.

2.3. *Stochastic control formulation*

For $s \in [0, T]$, let $\mathcal{U}_s$ denote the set of predictable U-valued control processes $\boldsymbol{u} = (\boldsymbol{u}_t)_{t \in [s,T]}$. The stochastic control problem associated with (7) is

$$v(s, \boldsymbol{x}) = \sup_{\boldsymbol{u} \in \mathcal{U}_s} J(s, \boldsymbol{x}; \boldsymbol{u}), \tag{8}$$

where the objective functional is defined by

$$J(s, \boldsymbol{x}; \boldsymbol{u}) = \mathbb{E}_{s,\boldsymbol{x}}\left[\int_s^T \mathrm{e}^{-\rho(t-s)} K \mathrm{e}^{-K\int_s^t u_r^\beta \,\mathrm{d}r} u_t^\beta g(t, \boldsymbol{X}_t^{\boldsymbol{u}})\,\mathrm{d}t + \mathrm{e}^{-\rho(T-s)} \mathrm{e}^{-K\int_s^T u_r^\beta \,\mathrm{d}r} g(T, \boldsymbol{X}_T^{\boldsymbol{u}})\right],$$

and the controlled state process $\boldsymbol{X}^{\boldsymbol{u}}$ satisfies

$$\mathrm{d}\boldsymbol{X}_t^{\boldsymbol{u}} = \boldsymbol{b}(t, \boldsymbol{X}_t^{\boldsymbol{u}}, \boldsymbol{u}_t^\alpha)\,\mathrm{d}t + \boldsymbol{\sigma}(t, \boldsymbol{X}_t^{\boldsymbol{u}}, \boldsymbol{u}_t^\alpha)\,\mathrm{d}\boldsymbol{W}_t + \int_{\mathbb{R}} \boldsymbol{\gamma}(t, \boldsymbol{X}_{t-}^{\boldsymbol{u}}, \boldsymbol{u}_t^\alpha, z)\, N(\mathrm{d}t, \mathrm{d}z), \quad \boldsymbol{X}_s^{\boldsymbol{u}} = \boldsymbol{x}. \tag{9}$$

To verify this representation, we first consider a general stochastic control problem with a control-

dependent discount factor:

$$\mathcal{V}(s,\boldsymbol{x}) := \sup_{\boldsymbol{u}\in\mathcal{U}_s} \bar{J}(s,\boldsymbol{x};\boldsymbol{u}), \tag{10}$$

where

$$\bar{J}(s,\boldsymbol{x};\boldsymbol{u}) = \mathbb{E}_{s,\boldsymbol{x}}\left[\int_s^T \mathrm{e}^{-\rho(t-s)}\mathrm{e}^{-\int_s^t f(r,\boldsymbol{X}_r^{\boldsymbol{u}},\boldsymbol{u}_r)\,\mathrm{d}r} G(t,\boldsymbol{X}_t^{\boldsymbol{u}},\boldsymbol{u}_t)\,\mathrm{d}t + \mathrm{e}^{-\rho(T-s)}\mathrm{e}^{-\int_s^T f(r,\boldsymbol{X}_r^{\boldsymbol{u}},\boldsymbol{u}_r)\,\mathrm{d}r} H(T,\boldsymbol{X}_T^{\boldsymbol{u}})\right].$$

Here, f, G, and H are continuous functions. Suppose that there exist constants $c, C > 0$ such that $f(t,\boldsymbol{x},\boldsymbol{u}) \geq -c$, $|G(t,\boldsymbol{x},\boldsymbol{u})| + |H(T,\boldsymbol{x})| \leq C\big(1+|\boldsymbol{x}|^p\big)$, for all $(t,\boldsymbol{x},\boldsymbol{u}) \in [0,T]\times\mathbb{R}^n\times U$, where p is the exponent in Assumption 2.1.

THEOREM 2.4 *Let $V \in C^{1,2}([0,T)\times\mathbb{R}^n)\cap C([0,T]\times\mathbb{R}^n)$ satisfy $|V(t,\boldsymbol{x})| \leq C_V(1+|\boldsymbol{x}|^p)$ for some constant $C_V > 0$. If V is a classical solution of the HJB equation*

$$\begin{cases} V_t - \rho V + \sup_{\boldsymbol{u}\in U}\big\{\mathcal{L}^{\boldsymbol{u}^\alpha} V + \mathcal{J}^{\boldsymbol{u}^\alpha} V - f(t,\boldsymbol{x},\boldsymbol{u})V + G(t,\boldsymbol{x},\boldsymbol{u})\big\} = 0, \quad (t,\boldsymbol{x}) \in [0,T)\times\mathbb{R}^n, \\ V(T,\boldsymbol{x}) = H(T,\boldsymbol{x}), \end{cases} \tag{11}$$

then $\mathcal{V}(s,\boldsymbol{x}) \leq V(s,\boldsymbol{x})$ for every $(s,\boldsymbol{x}) \in [0,T]\times\mathbb{R}^n$. Furthermore, suppose that a measurable selection $\boldsymbol{u}^*(t,\boldsymbol{x}) \in U$ attains the supremum in (11) at every $(t,\boldsymbol{x}) \in [0,T)\times\mathbb{R}^n$ and that the induced feedback control $\boldsymbol{u}_t^* = \boldsymbol{u}^*(t,\boldsymbol{X}_{t-}^{\boldsymbol{u}^*})$ belongs to $\mathcal{U}_s$. Then $V(s,\boldsymbol{x}) = \mathcal{V}(s,\boldsymbol{x}) = \bar{J}(s,\boldsymbol{x};\boldsymbol{u}^*)$ for every $(s,\boldsymbol{x}) \in [0,T]\times\mathbb{R}^n$.

The proof of Theorem 2.4 is given in Appendix A.2. For fixed $K > 0$, setting $f(t,\boldsymbol{x},\boldsymbol{u}) = Ku^\beta$, $G(t,\boldsymbol{x},\boldsymbol{u}) = Ku^\beta g(t,\boldsymbol{x})$ and $H(T,\boldsymbol{x}) = g(T,\boldsymbol{x})$ reduces $\bar{J}$ to the objective J in (8) and the HJB equation (11) to (7), yielding the following result.

COROLLARY 2.5 *Fix $K > 0$. Let $v \in C^{1,2}([0,T)\times\mathbb{R}^n)\cap C([0,T]\times\mathbb{R}^n)$ satisfy the polynomial-growth condition of Theorem 2.4 and be a classical solution of (7) with terminal condition $v(T,\boldsymbol{x}) = g(T,\boldsymbol{x})$. If the pointwise supremum in that equation admits a measurable maximizer whose induced feedback control belongs to $\mathcal{U}_s$, then v is the value function of (8), and the corresponding feedback control is optimal.*

Corollary 2.5 identifies the penalized optimal stopping problem as a stochastic control problem with a control-dependent discount factor. To incorporate this factor into the Markovian state, we follow Dai *et al.* (2026) and introduce the auxiliary process $R^{\boldsymbol{u}} = (R_t^{\boldsymbol{u}})_{t\in[s,T]}$, defined by

$$R_t^{\boldsymbol{u}} = r\exp\left(-K\int_s^t u_\ell^\beta\,\mathrm{d}\ell\right).$$

With the augmented state $(\boldsymbol{X}^{\boldsymbol{u}}, R^{\boldsymbol{u}})$, problem (8) becomes the Markovian control problem

$$\widetilde{v}(s,\boldsymbol{x},r) = \sup_{\boldsymbol{u}\in\mathcal{U}_s} \widetilde{J}(s,\boldsymbol{x},r;\boldsymbol{u}), \tag{12}$$

where the objective functional is

$$\widetilde{J}(s,\boldsymbol{x},r;\boldsymbol{u}) = \mathbb{E}_{s,\boldsymbol{x},r}\left[\int_s^T \mathrm{e}^{-\rho(\ell-s)} K R_\ell^{\boldsymbol{u}} u_\ell^\beta g(\ell,\boldsymbol{X}_\ell^{\boldsymbol{u}})\,\mathrm{d}\ell + \mathrm{e}^{-\rho(T-s)} R_T^{\boldsymbol{u}} g(T,\boldsymbol{X}_T^{\boldsymbol{u}})\right].$$

The linear dependence of $R^{\boldsymbol{u}}$ on its initial value r implies $\widetilde{J}(s,\boldsymbol{x},r;\boldsymbol{u}) = rJ(s,\boldsymbol{x};\boldsymbol{u})$ for every admissible control $\boldsymbol{u} \in \mathcal{U}_s$, and hence $\widetilde{v}(s,\boldsymbol{x},r) = rv(s,\boldsymbol{x})$. Setting $r = 1$ therefore recovers the original penalized value function: $\widetilde{v}(s,\boldsymbol{x},1) = v(s,\boldsymbol{x})$. Applying the DPP to the augmented control problem yields the following HJB equation:

$$\begin{cases} \widetilde{v}_t - \rho\widetilde{v} + \sup\limits_{\boldsymbol{u}\in U}\left\{\mathcal{L}^{\boldsymbol{u}^\alpha}\widetilde{v} + \mathcal{J}^{\boldsymbol{u}^\alpha}\widetilde{v} - Kru^\beta\widetilde{v}_r + Kru^\beta g(t,\boldsymbol{x})\right\} = 0, \quad (t,\boldsymbol{x},r) \in [0,T)\times\mathbb{R}^n\times(0,\infty), \\ \widetilde{v}(T,\boldsymbol{x},r) = rg(T,\boldsymbol{x}), \quad \widetilde{v}(t,\boldsymbol{x},0) = 0. \end{cases} \tag{13}$$

Substituting $\widetilde{v}(t,\boldsymbol{x},r) = rv(t,\boldsymbol{x})$ into (13) and dividing by $r > 0$ recovers (7). Thus, the penalized control problem (8), with its control-dependent discount factor, is equivalent to the augmented Markovian control problem (12). The next section introduces policy randomization and entropy regularization for this augmented penalized control problem.

3. Reinforcement learning formulation

Building on the penalty mixed control problem in Section 2, we introduce policy randomization and Shannon entropy regularization to formulate an exploratory control problem. Our approach draws on the continuous-time exploratory control framework of Wang *et al.* (2020), the randomized policies and policy improvement method of Jia and Zhou (2023), the action-sampling and relaxed-control formulations for jump processes in Bo *et al.* (2026), and the treatment of penalized optimal stopping in Dai *et al.* (2026). Each action combines a continuous-valued investment component with a binary stopping component, so randomized policies are defined on a mixed continuous–discrete action space. We first describe action sampling and its probability space, then formulate a relaxed exploratory control problem for dynamic programming, and finally present its HJB equation, Gibbs policy and policy improvement result.

Equip $U = A \times \{0,1\}$ with the reference measure $\mu_U := \mathrm{d}\boldsymbol{a} \otimes (\delta_0 + \delta_1)$, where $\mathrm{d}\boldsymbol{a}$ is Lebesgue measure on A and $\delta_0 + \delta_1$ is counting measure on $\{0,1\}$. Throughout, we write $\mathrm{d}\boldsymbol{u}$ for integration with respect to μ_U, so that, for any integrable F,

$$\int_U F(\boldsymbol{u})\,\mathrm{d}\boldsymbol{u} := \sum_{\beta\in\{0,1\}}\int_A F(\boldsymbol{a},\beta)\,\mathrm{d}\boldsymbol{a}.$$

Let $\mathcal{P}(U)$ denote the set of probability distributions on U that admit densities with respect to μ_U. For a time–state pair $(s,\boldsymbol{x})$, write $\pi(\mathrm{d}\boldsymbol{u} \mid s,\boldsymbol{x})$ for the conditional distribution of the mixed action and $\pi(\boldsymbol{u} \mid s,\boldsymbol{x})$ for its density with respect to μ_U. The set of admissible randomized feedback policies on $[t,T]$ is denoted by Π_t.

To sample actions, we enlarge the original probability space to support a sequence $(\xi_i)_{i\geq 0}$ of independent sampling variables, independent also of the Brownian motion $\boldsymbol{W}$ and the Poisson random measure N. Let Φ be a measurable sampling map such that $\Phi(s,\boldsymbol{x},\xi_i)$ has distribution $\pi(\mathrm{d}\boldsymbol{u} \mid s,\boldsymbol{x})$ for every $(s,\boldsymbol{x})$. We retain $\mathbb{P}$ for the original probability measure and denote the measure on the enlarged space by $\mathbb{Q}$, whose restriction to the original environmental $\boldsymbol{\sigma}$-algebra equals $\mathbb{P}$. Fix a time grid $\mathcal{G}_{t:T} = \{t = t_0 < t_1 < \cdots < t_N = T\}$. At each grid point t_i, sample an action conditional on the current state according to $\boldsymbol{u}_{t_i}^{\pi,\mathcal{G}} := \Phi(t_i, \boldsymbol{X}_{t_i}^{\pi,\mathcal{G}}, \xi_i)$ and hold it fixed on $[t_i, t_{i+1})$. Denote its investment and stopping components by $\boldsymbol{u}^{\alpha,\pi,\mathcal{G}}$ and $u^{\beta,\pi,\mathcal{G}}$, respectively, and suppress the superscript $\mathcal{G}$ when the grid is fixed. The enlarged filtration is $\mathbb{F}^{\mathbb{Q}} = (\mathcal{F}_s^{\mathbb{Q}})_{0\leq s\leq T}$, where $\mathcal{F}_s^{\mathbb{Q}} = \mathcal{F}_s^{\boldsymbol{W},N} \vee \boldsymbol{\sigma}(\xi_i : t_i \leq s)$. Unless otherwise specified, the expectations $\mathbb{E}_{s,\boldsymbol{x}}$ and $\mathbb{E}_{s,\boldsymbol{x},r}$ below are taken under $\mathbb{Q}$ with the indicated initial states.

For a policy $\pi \in \Pi_t$, a grid $\mathcal{G}_{t:T}$ and an initial time–state pair $(t,\boldsymbol{x}) \in [0,T]\times\mathbb{R}^n$, the sampled

wealth process $\boldsymbol{X}^\pi = \{\boldsymbol{X}_s^\pi : t \le s \le T\}$ satisfies

$$\mathrm{d}\boldsymbol{X}_s^\pi = \boldsymbol{b}(s, \boldsymbol{X}_s^\pi, \boldsymbol{u}_s^{\alpha,\pi})\,\mathrm{d}s + \boldsymbol{\sigma}(s, \boldsymbol{X}_s^\pi, \boldsymbol{u}_s^{\alpha,\pi})\,\mathrm{d}\boldsymbol{W}_s + \int_{\mathbb{R}} \boldsymbol{\gamma}(s, \boldsymbol{X}_{s-}^\pi, \boldsymbol{u}_s^{\alpha,\pi}, z)\, N(\mathrm{d}s, \mathrm{d}z), \quad \boldsymbol{X}_t^\pi = \boldsymbol{x}, \tag{14}$$

with the corresponding auxiliary process satisfying

$$\mathrm{d}R_s^\pi = -KR_s^\pi u_s^{\beta,\pi}\,\mathrm{d}s, \quad R_t^\pi = 1. \tag{15}$$

The state processes $\boldsymbol{X}^\pi$ and R^π, together with the sampled control $\boldsymbol{u}^\pi$, depend on the chosen grid; their grid superscripts are omitted only for notational convenience. Under the uniform conditions on the state coefficients in Assumption 2.1, (14) admits a unique strong solution for each fixed grid and satisfies the corresponding state moment bound. Together, $(\boldsymbol{X}^\pi, R^\pi)$ form the sampled augmented state process induced by action sampling on that grid.

The coefficients of the sampled augmented state process still depend on the exogenous action randomization. To obtain a Markovian control formulation with state dynamics determined by the current state and policy density, we use relaxed control to average over actions in the local characteristics of the state process and the running payoff. This construction follows the Markovian projection approach used in continuous-time exploratory control (Brunick and Shreve 2013, Jia and Zhou 2022a). For relaxed control with controlled jumps, see Kushner (2000); applications to continuous-time RL for jump-diffusions include Gao *et al.* (2026) and Bo *et al.* (2026).

Let $\bar{\boldsymbol{W}}$ be an n-dimensional Brownian motion on a suitably enlarged probability space, and define

$$\bar{\boldsymbol{b}}^\pi(s,\boldsymbol{x}) := \int_U \boldsymbol{b}(s,\boldsymbol{x},\boldsymbol{u}^\alpha)\pi(\boldsymbol{u} \mid s,\boldsymbol{x})\,\mathrm{d}\boldsymbol{u}, \quad \bar{\mathbf{a}}^\pi(s,\boldsymbol{x}) := \int_U \boldsymbol{\sigma}\boldsymbol{\sigma}^{\mathrm{T}}(s,\boldsymbol{x},\boldsymbol{u}^\alpha)\pi(\boldsymbol{u} \mid s,\boldsymbol{x})\,\mathrm{d}\boldsymbol{u},$$

with $\bar{\boldsymbol{\sigma}}^\pi(s,\boldsymbol{x}) := [\bar{\mathbf{a}}^\pi(s,\boldsymbol{x})]^{1/2}$. The relaxed wealth process $\bar{\boldsymbol{X}}^\pi$ and auxiliary process $\bar{R}^\pi$ satisfy

$$\mathrm{d}\bar{\boldsymbol{X}}_s^\pi = \bar{\boldsymbol{b}}^\pi(s, \bar{\boldsymbol{X}}_s^\pi)\,\mathrm{d}s + \bar{\boldsymbol{\sigma}}^\pi(s, \bar{\boldsymbol{X}}_s^\pi)\,\mathrm{d}\bar{\boldsymbol{W}}_s + \int_U \int_{\mathbb{R}} \boldsymbol{\gamma}(s, \bar{\boldsymbol{X}}_{s-}^\pi, \boldsymbol{u}^\alpha, z)\,\mathcal{N}^\pi(\mathrm{d}s, \mathrm{d}z, \mathrm{d}\boldsymbol{u}), \quad \bar{\boldsymbol{X}}_t^\pi = \boldsymbol{x}, \tag{16}$$

$$\mathrm{d}\bar{R}_s^\pi = -K\bar{R}_s^\pi \left(\int_U u^\beta \pi(\boldsymbol{u} \mid s, \bar{\boldsymbol{X}}_{s-}^\pi)\,\mathrm{d}\boldsymbol{u} \right) \mathrm{d}s, \quad \bar{R}_t^\pi = 1, \tag{17}$$

where $\mathcal{N}^\pi$ is a policy-dependent random measure with action marks and predictable compensator $\pi(\boldsymbol{u} \mid s, \bar{\boldsymbol{X}}_{s-}^\pi)\,\mathrm{d}\boldsymbol{u}\, m(\mathrm{d}z)\,\mathrm{d}s$. For controlled jump-diffusions, Theorem 2.3 in Bo *et al.* (2026) proves weak convergence of the grid-sampled state to its policy-averaged dynamics as the grid mesh tends to zero, under their stated regularity assumptions. In our problem, the same argument applies to the augmented state $(\boldsymbol{X}^\pi, R^\pi)$.

Definition 1 Fix $t \in [0,T]$. A policy $\pi = \pi(\cdot \mid \cdot,\cdot)$ is admissible on $[t,T]$, written $\pi \in \Pi_t$, if the following conditions hold:

(i) For every $(s,\boldsymbol{x}) \in [t,T] \times \mathbb{R}^n$, $\pi(\cdot \mid s,\boldsymbol{x})$ is a probability density with respect to μ_U, its associated distribution has support U, and $(s,\boldsymbol{x},\boldsymbol{u}) \mapsto \pi(\boldsymbol{u} \mid s,\boldsymbol{x})$ is measurable.

(ii) For each $\boldsymbol{u} \in U$, $\pi(\boldsymbol{u} \mid s,\boldsymbol{x})$ is continuous in $(s,\boldsymbol{x})$. The associated distributions are contin-

uous in total variation and uniformly Lipschitz in $\boldsymbol{x}$: there exists $C_\pi > 0$ such that

$$\int_U |\pi(\boldsymbol{u} \mid r, \boldsymbol{x}') - \pi(\boldsymbol{u} \mid s, \boldsymbol{x})| \,\mathrm{d}\boldsymbol{u} \longrightarrow 0, \quad (r, \boldsymbol{x}') \to (s, \boldsymbol{x}),$$
$$\int_U |\pi(\boldsymbol{u} \mid s, \boldsymbol{x}) - \pi(\boldsymbol{u} \mid s, \boldsymbol{x}')| \,\mathrm{d}\boldsymbol{u} \le C_\pi |\boldsymbol{x} - \boldsymbol{x}'|,$$

where the second relation holds for all $s, \boldsymbol{x}, \boldsymbol{x}'$.

(iii) The policy entropy $\mathcal{H}(\pi(\cdot \mid s, \boldsymbol{x})) := -\int_U \pi(\boldsymbol{u} \mid s, \boldsymbol{x}) \log \pi(\boldsymbol{u} \mid s, \boldsymbol{x}) \,\mathrm{d}\boldsymbol{u}$ is well defined. Moreover, there exists $C_\pi > 0$ such that, for all $s, \boldsymbol{x}, \boldsymbol{x}'$,

$$|\mathcal{H}(\pi(\cdot \mid s, \boldsymbol{x}))| \le C_\pi (1 + |\boldsymbol{x}|^p),$$
$$\int_U \int_{\mathbb{R}} |\boldsymbol{\gamma}(s, \boldsymbol{x}', \boldsymbol{u}^\alpha, z)| |\pi(\boldsymbol{u} \mid s, \boldsymbol{x}) - \pi(\boldsymbol{u} \mid s, \boldsymbol{x}')| \, m(\mathrm{d}z) \,\mathrm{d}\boldsymbol{u} \le C_\pi |\boldsymbol{x} - \boldsymbol{x}'|.$$

We further assume that, for every $\pi \in \Pi_s$, (16) admits a unique strong solution satisfying $\mathbb{E}_{s,\boldsymbol{x}}[\sup_{r \in [s,T]} |\bar{\boldsymbol{X}}_r^\pi|^p] \le C_{\pi,p,T}(1 + |\boldsymbol{x}|^p)$ for some constant $C_{\pi,p,T} > 0$.

To encourage exploration, we introduce Shannon entropy regularization. Following the treatment of penalized optimal stopping in Dai *et al.* (2026), we weight the entropy term by the auxiliary state process. This preserves the boundary condition $\widetilde{v}(t, \boldsymbol{x}, 0) = 0$ and the linear dependence of the augmented objective on the initial auxiliary state r. For the augmented problem, the dynamics in (17) start from $\bar{R}_s^\pi = r$, and the entropy-regularized exploratory control problem is

$$\begin{aligned}
\widetilde{v}^\Pi(s, \boldsymbol{x}, r) &= \sup_{\pi \in \Pi_s} \widetilde{J}^\pi(s, \boldsymbol{x}, r) \\
&= \sup_{\pi \in \Pi_s} \mathbb{E}_{s,\boldsymbol{x},r}\Bigg[\int_s^T \mathrm{e}^{-\rho(\ell - s)} \bar{R}_\ell^\pi \Bigg(K g(\ell, \bar{\boldsymbol{X}}_\ell^\pi) \int_U u^\beta \pi(\boldsymbol{u} \mid \ell, \bar{\boldsymbol{X}}_\ell^\pi) \,\mathrm{d}\boldsymbol{u} \\
&\qquad + \lambda \mathcal{H}\big(\pi(\cdot \mid \ell, \bar{\boldsymbol{X}}_\ell^\pi)\big) \Bigg) \mathrm{d}\ell + \mathrm{e}^{-\rho(T-s)} \bar{R}_T^\pi g(T, \bar{\boldsymbol{X}}_T^\pi) \Bigg],
\end{aligned} \tag{18}$$

where $\lambda > 0$ is the temperature parameter. Since $\bar{R}^\pi$ is linear in r, $\widetilde{J}^\pi(s, \boldsymbol{x}, r) = r J^\pi(s, \boldsymbol{x})$ for every policy π, and hence $\widetilde{v}^\Pi(s, \boldsymbol{x}, r) = r v^\Pi(s, \boldsymbol{x})$. In particular, $\widetilde{v}^\Pi(s, \boldsymbol{x}, 1) = v^\Pi(s, \boldsymbol{x})$, so the augmented problem is equivalent to the reduced exploratory control problem

$$\begin{aligned}
v^\Pi(s, \boldsymbol{x}) &= \sup_{\pi \in \Pi_s} J^\pi(s, \boldsymbol{x}) \\
&= \sup_{\pi \in \Pi_s} \mathbb{E}_{s,\boldsymbol{x}}\Bigg[\int_s^T \mathrm{e}^{-\rho(\ell - s) - K \int_s^\ell \int_U u^\beta \pi(\boldsymbol{u} \mid \xi, \bar{\boldsymbol{X}}_\xi^\pi) \,\mathrm{d}\boldsymbol{u} \,\mathrm{d}\xi} \Bigg(K g(\ell, \bar{\boldsymbol{X}}_\ell^\pi) \int_U u^\beta \pi(\boldsymbol{u} \mid \ell, \bar{\boldsymbol{X}}_\ell^\pi) \,\mathrm{d}\boldsymbol{u} \\
&\qquad + \lambda \mathcal{H}\big(\pi(\cdot \mid \ell, \bar{\boldsymbol{X}}_\ell^\pi)\big) \Bigg) \mathrm{d}\ell + \mathrm{e}^{-\rho(T-s) - K \int_s^T \int_U u^\beta \pi(\boldsymbol{u} \mid \xi, \bar{\boldsymbol{X}}_\xi^\pi) \,\mathrm{d}\boldsymbol{u} \,\mathrm{d}\xi} g(T, \bar{\boldsymbol{X}}_T^\pi) \Bigg].
\end{aligned} \tag{19}$$

By Theorem 2.4, the associated HJB equation is

$$\begin{cases}
v_t^\Pi(t, \boldsymbol{x}) - \rho v^\Pi(t, \boldsymbol{x}) + \sup\limits_{\pi \in \mathcal{P}(U)} \displaystyle\int_U \Big[\mathcal{L}^{\boldsymbol{u}^\alpha} v^\Pi + \mathcal{J}^{\boldsymbol{u}^\alpha} v^\Pi + K u^\beta (g - v^\Pi) - \lambda \log \pi(\boldsymbol{u}) \Big] \pi(\boldsymbol{u}) \,\mathrm{d}\boldsymbol{u} = 0, \\
v^\Pi(T, \boldsymbol{x}) = g(T, \boldsymbol{x}),
\end{cases} \tag{20}$$

and applying the verification argument of Theorem 2.4 to the relaxed jump-diffusion state process in (19) yields the following result.

COROLLARY 3.1 *Let $v \in C^{1,2}([0,T)\times\mathbb{R}^n)\cap C([0,T]\times\mathbb{R}^n)$ satisfy $|v(t,\boldsymbol{x})| \le C_v(1+|\boldsymbol{x}|^p)$ and be a classical solution of* (20) *with terminal condition $v(T,\boldsymbol{x}) = g(T,\boldsymbol{x})$. If the pointwise supremum in the HJB equation is attained by a Gibbs policy in the admissible policy class, then v is the value function of* (19)*, and that Gibbs policy is optimal.*

For notational convenience, define the Hamiltonian

$$H(t,\boldsymbol{x},\boldsymbol{u};\varphi) := \mathcal{L}^{\boldsymbol{u}^\alpha}\varphi(t,\boldsymbol{x}) + \mathcal{J}^{\boldsymbol{u}^\alpha}\varphi(t,\boldsymbol{x}) + Ku^\beta\big(g(t,\boldsymbol{x}) - \varphi(t,\boldsymbol{x})\big).$$

For each fixed $(t,\boldsymbol{x})$, the pointwise maximization in (20) is the entropy-regularized problem $\sup_{\pi\in\mathcal{P}(U)}\int_U [H(t,\boldsymbol{x},\boldsymbol{u};v^\Pi) - \lambda\log\pi(\boldsymbol{u})]\pi(\boldsymbol{u})\,\mathrm{d}\boldsymbol{u}$. For $\lambda > 0$, the first-order optimality condition under the constraint $\int_U \pi(\boldsymbol{u})\,\mathrm{d}\boldsymbol{u} = 1$ yields the Gibbs policy

$$\pi^*(\boldsymbol{u}\mid t,\boldsymbol{x}) = \frac{\exp\big(H(t,\boldsymbol{x},\boldsymbol{u};v^\Pi)/\lambda\big)}{\displaystyle\int_U \exp\big(H(t,\boldsymbol{x},\bar{\boldsymbol{u}};v^\Pi)/\lambda\big)\,\mathrm{d}\bar{\boldsymbol{u}}}. \tag{21}$$

The corresponding HJB equation for the augmented exploratory problem is

$$\begin{cases} \widetilde{v}^\Pi_t - \rho\widetilde{v}^\Pi + \sup\limits_{\pi\in\mathcal{P}(U)}\displaystyle\int_U \Big[\mathcal{L}^{\boldsymbol{u}^\alpha}\widetilde{v}^\Pi + \mathcal{J}^{\boldsymbol{u}^\alpha}\widetilde{v}^\Pi - Kru^\beta\widetilde{v}^\Pi_r + Kru^\beta g - r\lambda\log\pi(\boldsymbol{u})\Big]\pi(\boldsymbol{u})\,\mathrm{d}\boldsymbol{u} = 0, \\ \widetilde{v}^\Pi(T,\boldsymbol{x},r) = rg(T,\boldsymbol{x}), \quad \widetilde{v}^\Pi(t,\boldsymbol{x},0) = 0. \end{cases} \tag{22}$$

Defining the augmented Hamiltonian by

$$\widetilde{H}(t,\boldsymbol{x},r,\boldsymbol{u};\widetilde{v}) := \mathcal{L}^{\boldsymbol{u}^\alpha}\widetilde{v} + \mathcal{J}^{\boldsymbol{u}^\alpha}\widetilde{v} - Kru^\beta\widetilde{v}_r + Kru^\beta g,$$

the relation $\widetilde{v}^\Pi(t,\boldsymbol{x},r) = rv^\Pi(t,\boldsymbol{x})$ implies $\widetilde{H}(t,\boldsymbol{x},r,\boldsymbol{u};\widetilde{v}^\Pi) = rH(t,\boldsymbol{x},\boldsymbol{u};v^\Pi)$. Hence, for $r > 0$, the optimal policy for the augmented problem is

$$\widetilde{\pi}^*(\boldsymbol{u}\mid t,\boldsymbol{x},r) = \frac{\exp\Big(\widetilde{H}(t,\boldsymbol{x},r,\boldsymbol{u};\widetilde{v}^\Pi)/(r\lambda)\Big)}{\displaystyle\int_U \exp\Big(\widetilde{H}(t,\boldsymbol{x},r,\bar{\boldsymbol{u}};\widetilde{v}^\Pi)/(r\lambda)\Big)\,\mathrm{d}\bar{\boldsymbol{u}}} = \pi^*(\boldsymbol{u}\mid t,\boldsymbol{x}).$$

The augmented and reduced exploratory problems therefore induce the same optimal policy, allowing the subsequent algorithm to use the standard control structure of the augmented problem while learning only the reduced value function.

Before presenting policy improvement, consider an arbitrary admissible randomized policy $\pi\in\Pi_t$ whose value function satisfies $J^\pi \in C^{1,2}([0,T)\times\mathbb{R}^n)\cap C([0,T]\times\mathbb{R}^n)$. The DPP for the fixed policy gives the policy evaluation equation

$$\begin{cases} J^\pi_t(t,\boldsymbol{x}) - \rho J^\pi(t,\boldsymbol{x}) + \displaystyle\int_U \left[H(t,\boldsymbol{x},\boldsymbol{u};J^\pi) - \lambda\log\pi(\boldsymbol{u}\mid t,\boldsymbol{x})\right]\pi(\boldsymbol{u}\mid t,\boldsymbol{x})\,\mathrm{d}\boldsymbol{u} = 0, \\ J^\pi(T,\boldsymbol{x}) = g(T,\boldsymbol{x}). \end{cases} \tag{23}$$

THEOREM 3.2 *Fix $\pi \in \Pi_t$ and suppose that J^π has the classical regularity stated above. Define*

$$\mathcal{I}(\pi)(\boldsymbol{u} \mid s, \boldsymbol{x}) := \frac{\exp(H(s, \boldsymbol{x}, \boldsymbol{u}; J^\pi)/\lambda)}{\displaystyle\int_U \exp(H(s, \boldsymbol{x}, \bar{\boldsymbol{u}}; J^\pi)/\lambda)\, \mathrm{d}\bar{\boldsymbol{u}}}. \tag{24}$$

If $\pi^{\text{new}} := \mathcal{I}(\pi) \in \Pi_t$, then $J^{\pi^{\text{new}}}(s, \boldsymbol{x}) \geq J^\pi(s, \boldsymbol{x})$ for every $(s, \boldsymbol{x}) \in [t, T] \times \mathbb{R}^n$. Furthermore, if there exists a fixed point $\pi^* \in \Pi_t$ such that $\mathcal{I}(\pi^*) = \pi^*$ and J^{π^*} has the same classical regularity, then π^* is optimal for (19)*, and J^{π^*} is its value function.*

The proof of Theorem 3.2 is provided in Appendix A.3. Once the value function J^π associated with the current policy π has been evaluated, the Gibbs formula yields an improved policy. To implement this policy-evaluation and improvement procedure using sample-path observations, the next section introduces an action-dependent continuous-time q-function.

4. Continuous-time q-function and martingale characterizations

Building on the entropy-regularized exploratory problem, we introduce a continuous-time q-function for the mixed-control optimal stopping problem with jumps. Following the short-time action perturbation of Jia and Zhou (2023), we hold an action fixed over a short interval and expand the resulting $Q_{\Delta t}$-function around the policy value function. The coefficient of the first-order term in Δt defines the continuous-time q-function. For the present jump-diffusion model, the expansion also includes the nonlocal jump generator, following the related constructions of Gao *et al.* (2026) and Bo *et al.* (2026). The resulting q-function underpins the martingale characterizations and learning algorithm developed below.

4.1. *Q-function*

Fix a policy $\pi \in \Pi_t$, an initial time–state pair $(t, \boldsymbol{x}) \in [0, T) \times \mathbb{R}^n$ and a mixed action $\boldsymbol{u} = (\boldsymbol{u}^\alpha, u^\beta) \in U$. Following the short-time action perturbation in Jia and Zhou (2023), we hold $\boldsymbol{u}$ fixed on $[t, t + \Delta t)$ and sample actions according to π from $t + \Delta t$ onward, where $0 < \Delta t \leq T - t$. During the fixed-action interval, the wealth process satisfies

$$\mathrm{d}\boldsymbol{X}_s^{\boldsymbol{u}} = \boldsymbol{b}(s, \boldsymbol{X}_s^{\boldsymbol{u}}, \boldsymbol{u}^\alpha)\, \mathrm{d}s + \boldsymbol{\sigma}(s, \boldsymbol{X}_s^{\boldsymbol{u}}, \boldsymbol{u}^\alpha)\, \mathrm{d}\boldsymbol{W}_s + \int_{\mathbb{R}} \boldsymbol{\gamma}(s, \boldsymbol{X}_{s-}^{\boldsymbol{u}}, \boldsymbol{u}^\alpha, z)\, N(\mathrm{d}s, \mathrm{d}z), \quad \boldsymbol{X}_t^{\boldsymbol{u}} = \boldsymbol{x}.$$

For fixed $\Delta t > 0$, the expected payoff generated by the perturbed policy is

$$
\begin{aligned}
&Q_{\Delta t}(t,\boldsymbol{x},\boldsymbol{u};\pi)\\
&=\mathbb{E}_{t,\boldsymbol{x}}\Bigg[\int_t^{t+\Delta t}\mathrm{e}^{-(\rho+Ku^\beta)(s-t)}Ku^\beta g(s,\boldsymbol{X}_s^{\boldsymbol{u}})\,\mathrm{d}s\\
&\quad+\mathrm{e}^{-(\rho+Ku^\beta)\Delta t}\mathbb{E}\Bigg[\int_{t+\Delta t}^{T}\mathrm{e}^{-\rho(s-t-\Delta t)-K\int_{t+\Delta t}^{s}u_v^{\beta,\pi}\,\mathrm{d}v}\Big(Ku_s^{\beta,\pi}g(s,\boldsymbol{X}_s^{\pi})-\lambda\log\pi(\boldsymbol{u}_s^{\pi}\mid s,\boldsymbol{X}_{s-}^{\pi})\Big)\,\mathrm{d}s\\
&\quad+\mathrm{e}^{-\rho(T-t-\Delta t)-K\int_{t+\Delta t}^{T}u_v^{\beta,\pi}\,\mathrm{d}v}g(T,\boldsymbol{X}_T^{\pi})\;\Big|\;\mathcal{F}_{t+\Delta t}^{\mathbb{Q}}\Bigg]\Bigg]\\
&=J^\pi(t,\boldsymbol{x})+\mathbb{E}_{t,\boldsymbol{x}}\Bigg[\int_t^{t+\Delta t}\mathrm{e}^{-(\rho+Ku^\beta)(s-t)}\big[J_s^\pi(s,\boldsymbol{X}_s^{\boldsymbol{u}})+H(s,\boldsymbol{X}_s^{\boldsymbol{u}},\boldsymbol{u};J^\pi)-\rho J^\pi(s,\boldsymbol{X}_s^{\boldsymbol{u}})\big]\,\mathrm{d}s\Bigg]\\
&=J^\pi(t,\boldsymbol{x})+\big[J_t^\pi(t,\boldsymbol{x})+H(t,\boldsymbol{x},\boldsymbol{u};J^\pi)-\rho J^\pi(t,\boldsymbol{x})\big]\Delta t+o(\Delta t).
\end{aligned}
\tag{25}
$$

By construction, no policy-entropy term is charged over the fixed-action interval, while entropy rewards from $t+\Delta t$ onward are included in the continuation value. The zeroth-order term is thus $J^\pi(t,\boldsymbol{x})$, independent of $\boldsymbol{u}$, while the current action affects the first-order term.

4.2. *Continuous-time q-function*

The expansion in (25) shows that the first-order deviation of $Q_{\Delta t}$ from J^π measures the local value contribution of action $\boldsymbol{u}$. This motivates the following definition for the entropy-regularized mixed-control exploratory problem (19), with a fixed penalty parameter $K>0$ and policy π.

Definition 2 For $(t,\boldsymbol{x},\boldsymbol{u})\in[0,T]\times\mathbb{R}^n\times U$, define the continuous-time q-function by

$$
q(t,\boldsymbol{x},\boldsymbol{u};\pi):=J_t^\pi(t,\boldsymbol{x})+\mathcal{L}^{\boldsymbol{u}^\alpha}J^\pi(t,\boldsymbol{x})+\mathcal{J}^{\boldsymbol{u}^\alpha}J^\pi(t,\boldsymbol{x})-\rho J^\pi(t,\boldsymbol{x})+Ku^\beta\big(g(t,\boldsymbol{x})-J^\pi(t,\boldsymbol{x})\big).
\tag{26}
$$

By (25), we have

$$
q(t,\boldsymbol{x},\boldsymbol{u};\pi)=\lim_{\Delta t\downarrow 0}\frac{Q_{\Delta t}(t,\boldsymbol{x},\boldsymbol{u};\pi)-J^\pi(t,\boldsymbol{x})}{\Delta t}.
$$

Thus, $q(t,\boldsymbol{x},\boldsymbol{u};\pi)$ is the instantaneous value contribution of holding action $\boldsymbol{u}$ fixed relative to the current policy value function. The term $\mathcal{L}^{\boldsymbol{u}^\alpha}J^\pi+\mathcal{J}^{\boldsymbol{u}^\alpha}J^\pi$ captures the effect of the investment action on wealth dynamics, $-\rho J^\pi$ accounts for discounting, and $Ku^\beta(g-J^\pi)$ measures the difference between the stopping payoff and continuation value.

By Theorem 3.2, once $q(\cdot,\cdot,\cdot;\pi)$ has been learned for a given policy, an improved policy can be constructed through the Gibbs formula

$$
\pi^{\mathrm{new}}(\boldsymbol{u}\mid t,\boldsymbol{x})=\frac{\exp(q(t,\boldsymbol{x},\boldsymbol{u};\pi)/\lambda)}{\displaystyle\int_U\exp(q(t,\boldsymbol{x},\bar{\boldsymbol{u}};\pi)/\lambda)\,\mathrm{d}\bar{\boldsymbol{u}}}.
$$

4.3. *Martingale characterizations*

We establish martingale characterizations of the q-function and of the value function–q-function .

THEOREM 4.1 *Fix $\pi \in \Pi_t$ with value function $J^\pi \in C^{1,2}([0,T)\times\mathbb{R}^n)\cap C([0,T]\times\mathbb{R}^n)$, and let $\widehat{q}:[0,T]\times\mathbb{R}^n\times U\to\mathbb{R}$ be continuous. In each assertion below, the state pair solves (14)–(15) under the indicated policy, starting from $(\boldsymbol{x},r)$ at time t. The following statements hold:*

(i) The identity $\widehat{q}(t,\boldsymbol{x},\boldsymbol{u}) = q(t,\boldsymbol{x},\boldsymbol{u};\pi)$ holds for every $(t,\boldsymbol{x},\boldsymbol{u})\in[0,T]\times\mathbb{R}^n\times U$ if and only if, for every initial pair $(t,\boldsymbol{x})\in[0,T]\times\mathbb{R}^n$ and time grid $\mathcal{G}_{t:T}$, the process

$$\begin{aligned} M_s :=\ & \mathrm{e}^{-\rho(s-t)}R_s^\pi J^\pi(s,\boldsymbol{X}_s^\pi) \\ & + \int_t^s \mathrm{e}^{-\rho(r-t)}R_r^\pi\left[Ku_r^{\beta,\pi}g(r,\boldsymbol{X}_r^\pi) - \widehat{q}(r,\boldsymbol{X}_r^\pi,\boldsymbol{u}_r^\pi)\right]\mathrm{d}r, \quad s\in[t,T], \end{aligned} \tag{27}$$

is an $(\mathbb{F}^\mathbb{Q},\mathbb{Q})$-martingale.

(ii) If $\widehat{q}(t,\boldsymbol{x},\boldsymbol{u}) = q(t,\boldsymbol{x},\boldsymbol{u};\pi)$ for every $(t,\boldsymbol{x},\boldsymbol{u})\in[0,T]\times\mathbb{R}^n\times U$, then, for every behavior policy $\pi'\in\Pi_t$, initial pair $(t,\boldsymbol{x})\in[0,T]\times\mathbb{R}^n$ and time grid $\mathcal{G}_{t:T}$, the process

$$\begin{aligned} M_s' :=\ & \mathrm{e}^{-\rho(s-t)}R_s^{\pi'}J^\pi(s,\boldsymbol{X}_s^{\pi'}) \\ & + \int_t^s \mathrm{e}^{-\rho(r-t)}R_r^{\pi'}\left[Ku_r^{\beta,\pi'}g(r,\boldsymbol{X}_r^{\pi'}) - \widehat{q}(r,\boldsymbol{X}_r^{\pi'},\boldsymbol{u}_r^{\pi'})\right]\mathrm{d}r, \quad s\in[t,T], \end{aligned} \tag{28}$$

is an $(\mathbb{F}^\mathbb{Q},\mathbb{Q})$-martingale.

(iii) If there exists a behavior policy $\pi'\in\Pi_t$ for which (28) is an $(\mathbb{F}^\mathbb{Q},\mathbb{Q})$-martingale for every initial pair $(t,\boldsymbol{x})\in[0,T]\times\mathbb{R}^n$ and time grid $\mathcal{G}_{t:T}$, then $\widehat{q}(t,\boldsymbol{x},\boldsymbol{u}) = q(t,\boldsymbol{x},\boldsymbol{u};\pi)$ for every $(t,\boldsymbol{x},\boldsymbol{u})\in[0,T]\times\mathbb{R}^n\times U$.

In each case, the q-function also satisfies

$$\int_U \left[q(t,\boldsymbol{x},\boldsymbol{u};\pi) - \lambda\log\pi(\boldsymbol{u}\mid t,\boldsymbol{x})\right]\pi(\boldsymbol{u}\mid t,\boldsymbol{x})\,\mathrm{d}\boldsymbol{u} = 0, \quad (t,\boldsymbol{x})\in[0,T]\times\mathbb{R}^n. \tag{29}$$

THEOREM 4.2 *Fix $\pi\in\Pi_t$. Let $\widehat{J}\in C^{1,2}([0,T)\times\mathbb{R}^n)\cap C([0,T]\times\mathbb{R}^n)$ satisfy $|\widehat{J}(s,\boldsymbol{x})|\le C_{\widehat{J}}(1+|\boldsymbol{x}|^p)$, and let $\widehat{q}:[0,T]\times\mathbb{R}^n\times U\to\mathbb{R}$ be continuous. Suppose that*

$$\begin{aligned} \widehat{J}(T,\boldsymbol{x}) &= g(T,\boldsymbol{x}), \\ \int_U \left[\widehat{q}(s,\boldsymbol{x},\boldsymbol{u}) - \lambda\log\pi(\boldsymbol{u}\mid s,\boldsymbol{x})\right]\pi(\boldsymbol{u}\mid s,\boldsymbol{x})\,\mathrm{d}\boldsymbol{u} &= 0, \quad (s,\boldsymbol{x})\in[0,T]\times\mathbb{R}^n. \end{aligned} \tag{30}$$

In each assertion below, the state pair solves (14)–(15) under the indicated policy, starting from $(\boldsymbol{x},r)$ at time t. The following statements hold:

(i) The functions $\widehat{J}$ and $\widehat{q}$ are the value function and q-function of π, respectively, if and only if, for every initial pair $(t,\boldsymbol{x})\in[0,T]\times\mathbb{R}^n$ and time grid $\mathcal{G}_{t:T}$, the process

$$\begin{aligned} \widehat{M}_s :=\ & \mathrm{e}^{-\rho(s-t)}R_s^\pi\widehat{J}(s,\boldsymbol{X}_s^\pi) \\ & + \int_t^s \mathrm{e}^{-\rho(r-t)}R_r^\pi\left[Ku_r^{\beta,\pi}g(r,\boldsymbol{X}_r^\pi) - \widehat{q}(r,\boldsymbol{X}_r^\pi,\boldsymbol{u}_r^\pi)\right]\mathrm{d}r, \quad s\in[t,T], \end{aligned} \tag{31}$$

is an $(\mathbb{F}^\mathbb{Q},\mathbb{Q})$-martingale.

(ii) If $\widehat{J}$ and $\widehat{q}$ are the value function and q-function of π, respectively, then, for every behavior

policy $\pi' \in \Pi_t$*, initial pair* $(t, \boldsymbol{x}) \in [0,T] \times \mathbb{R}^n$ *and time grid* $\mathcal{G}_{t:T}$*, the process*

$$\begin{aligned}\widehat{M}'_s &:= \mathrm{e}^{-\rho(s-t)} R_s^{\pi'} \widehat{J}(s, \boldsymbol{X}_s^{\pi'}) \\ &\quad + \int_t^s \mathrm{e}^{-\rho(r-t)} R_r^{\pi'} \left[K u_r^{\beta,\pi'} g(r, \boldsymbol{X}_r^{\pi'}) - \widehat{q}(r, \boldsymbol{X}_r^{\pi'}, \boldsymbol{u}_r^{\pi'}) \right] \mathrm{d}r, \quad s \in [t,T],\end{aligned} \tag{32}$$

is an $(\mathbb{F}^{\mathbb{Q}}, \mathbb{Q})$*-martingale.*

(iii) If there exists a behavior policy $\pi' \in \Pi_t$ *for which* (32) *is an* $(\mathbb{F}^{\mathbb{Q}}, \mathbb{Q})$*-martingale for every initial pair* $(t, \boldsymbol{x}) \in [0,T] \times \mathbb{R}^n$ *and time grid* $\mathcal{G}_{t:T}$*, then* $\widehat{J}$ *and* $\widehat{q}$ *are the value function and q-function of* π*, respectively.*

In each case, if $\pi(\boldsymbol{u} \mid t, \boldsymbol{x}) = \frac{\exp(\widehat{q}(t,\boldsymbol{x},\boldsymbol{u})/\lambda)}{\int_U \exp(\widehat{q}(t,\boldsymbol{x},\bar{\boldsymbol{u}})/\lambda)\, \mathrm{d}\bar{\boldsymbol{u}}}$*, then* π *is optimal with value function* $\widehat{J}$*.*

Theorem 4.2 characterizes the value function and q-function jointly through a single martingale condition, providing the theoretical basis for the q-learning algorithms below. The terminal condition in (30) specifies the correct terminal payoff, while the Shannon entropy-consistency condition, together with the martingale condition, identifies the q-function under π. The martingale condition alone is generally insufficient to identify $(\widehat{J}, \widehat{q})$ uniquely; the terminal and entropy-consistency conditions are also needed. In Theorem 4.1, by contrast, these constraints are already embodied in the prescribed policy value function J^π.

Together, Theorems 4.1 and 4.2 establish policy evaluation for continuous-time q-learning. Combined with Theorem 3.2, they yield an iterative scheme: learn (J^{π_k}, q^{π_k}) under π_k, then improve the policy using the Gibbs rule induced by q^{π_k}. The next section develops structured q-learning algorithms within this framework of martingale-based evaluation and Gibbs policy improvement.

5. Structured q-learning under Shannon entropy regularization

Building on the martingale characterizations of the continuous-time q-function in Section 4, we develop a structured q-learning algorithm for the mixed control problem using the parameter-update scheme of Jia and Zhou (2023). We approximate the policy value function by $V_{\boldsymbol{\theta}}$ and construct $q_{\boldsymbol{\theta},\boldsymbol{\varphi}}$ from the penalty HJB equation rather than introduce an independent q-function approximator $q_{\boldsymbol{\psi}}$. The unknown market parameters $\boldsymbol{\varphi}$ enter through the controlled generator and are learned jointly with the value-function parameters. Specifically, the structured q-function is

$$q_{\boldsymbol{\theta},\boldsymbol{\varphi}}(t,\boldsymbol{x},\boldsymbol{u}) := \partial_t V_{\boldsymbol{\theta}}(t,\boldsymbol{x}) + \mathcal{L}_{\boldsymbol{\varphi}}^{\boldsymbol{u}^\alpha} V_{\boldsymbol{\theta}}(t,\boldsymbol{x}) + \mathcal{J}_{\boldsymbol{\varphi}}^{\boldsymbol{u}^\alpha} V_{\boldsymbol{\theta}}(t,\boldsymbol{x}) - \rho V_{\boldsymbol{\theta}}(t,\boldsymbol{x}) + K u^\beta \big(g(t,\boldsymbol{x}) - V_{\boldsymbol{\theta}}(t,\boldsymbol{x})\big), \tag{33}$$

and, for given $(\boldsymbol{\theta}, \boldsymbol{\varphi})$, its induced Gibbs policy is

$$\pi_{\boldsymbol{\theta},\boldsymbol{\varphi}}(\boldsymbol{u} \mid t, \boldsymbol{x}) := \frac{\exp(q_{\boldsymbol{\theta},\boldsymbol{\varphi}}(t,\boldsymbol{x},\boldsymbol{u})/\lambda)}{\displaystyle\int_U \exp(q_{\boldsymbol{\theta},\boldsymbol{\varphi}}(t,\boldsymbol{x},\bar{\boldsymbol{u}})/\lambda)\, \mathrm{d}\bar{\boldsymbol{u}}}. \tag{34}$$

Set $t_l = l\Delta t$, $l = 0, \ldots, L$, and write $\boldsymbol{X}_l := \boldsymbol{X}_{t_l}$ and $R_l := R_{t_l}$. At step l, sample $\boldsymbol{u}_l \sim \pi_{\boldsymbol{\theta},\boldsymbol{\varphi}}(\cdot \mid t_l, \boldsymbol{X}_l)$ from the current policy. Given $\boldsymbol{u}_l$, the environment simulator generates $\boldsymbol{X}_{l+1}$ by an Euler discretization of the controlled wealth dynamics using the true market parameters, which are unknown to the learner. The parameter $\boldsymbol{\varphi}$ is used to construct the structured q-function and is updated from observed sample paths. The one-step wealth update is

$$\boldsymbol{X}_{l+1} = \boldsymbol{X}_l + \boldsymbol{b}(t_l, \boldsymbol{X}_l, \boldsymbol{u}_l^\alpha)\Delta t + \boldsymbol{\sigma}(t_l, \boldsymbol{X}_l, \boldsymbol{u}_l^\alpha)\Delta \boldsymbol{W}_l + \int_{\mathbb{R}} \boldsymbol{\gamma}(t_l, \boldsymbol{X}_l, \boldsymbol{u}_l^\alpha, z) N((t_l, t_{l+1}], \mathrm{d}z), \tag{35}$$

where $\Delta \boldsymbol{W}_l = \sqrt{\Delta t}\,\boldsymbol{\varepsilon}_l$, $\boldsymbol{\varepsilon}_l \sim N(0, \mathbf{I}_d)$, and $N((t_l, t_{l+1}] \times B) \sim \text{Poisson}(\Delta t\, m(B))$ for every $B \in \mathcal{B}(\mathbb{R})$. The corresponding auxiliary-state update is

$$R_{l+1} = R_l \exp(-K u_l^{\beta} \Delta t), \quad R_0 = 1. \tag{36}$$

By Theorem 4.2, when $(V_{\boldsymbol{\theta}}, q_{\boldsymbol{\theta},\boldsymbol{\varphi}})$ represents the value function and q-function of the current policy, the associated discounted reward process satisfies the martingale condition. Discretizing this condition on $\{t_l\}_{l=0}^{L}$ gives the one-step martingale increment

$$\Delta_l^{\boldsymbol{\theta},\boldsymbol{\varphi}} := \mathrm{e}^{-\rho t_{l+1}} R_{l+1} V_{\boldsymbol{\theta}}(t_{l+1}, \boldsymbol{X}_{l+1}) - \mathrm{e}^{-\rho t_l} R_l V_{\boldsymbol{\theta}}(t_l, \boldsymbol{X}_l) + \mathrm{e}^{-\rho t_l} R_l \left[K u_l^{\beta} g(t_l, \boldsymbol{X}_l) - q_{\boldsymbol{\theta},\boldsymbol{\varphi}}(t_l, \boldsymbol{X}_l, \boldsymbol{u}_l) \right] \Delta t, \tag{37}$$

and, for test processes $(\boldsymbol{\xi}_l^{\boldsymbol{\theta}})_{l=0}^{L-1}$ and $(\boldsymbol{\xi}_l^{\boldsymbol{\varphi}})_{l=0}^{L-1}$ adapted to the discrete-time filtration, the corresponding discrete martingale orthogonality conditions are

$$\mathbb{E}\left[\sum_{l=0}^{L-1} \boldsymbol{\xi}_l^{\boldsymbol{\theta}} \Delta_l^{\boldsymbol{\theta},\boldsymbol{\varphi}}\right] = 0, \qquad \mathbb{E}\left[\sum_{l=0}^{L-1} \boldsymbol{\xi}_l^{\boldsymbol{\varphi}} \Delta_l^{\boldsymbol{\theta},\boldsymbol{\varphi}}\right] = 0,$$

where we take $\boldsymbol{\xi}_l^{\boldsymbol{\theta}} = \mathrm{e}^{-\rho t_l} R_l \nabla_{\boldsymbol{\theta}} V_{\boldsymbol{\theta}}(t_l, \boldsymbol{X}_l)$ and $\boldsymbol{\xi}_l^{\boldsymbol{\varphi}} = \nabla_{\boldsymbol{\varphi}} q_{\boldsymbol{\theta},\boldsymbol{\varphi}}(t_l, \boldsymbol{X}_l, \boldsymbol{u}_l)$. Write $\boldsymbol{S}_{\boldsymbol{\theta}} := \sum_{l=0}^{L-1} \boldsymbol{\xi}_l^{\boldsymbol{\theta}} \Delta_l^{\boldsymbol{\theta},\boldsymbol{\varphi}}$ and $\boldsymbol{S}_{\boldsymbol{\varphi}} := \sum_{l=0}^{L-1} \boldsymbol{\xi}_l^{\boldsymbol{\varphi}} \Delta_l^{\boldsymbol{\theta},\boldsymbol{\varphi}}$ for the full-trajectory update quantities.

Joint identification in Theorem 4.2 also requires the terminal and entropy-consistency conditions. Jia and Zhou (2023) impose these conditions through the parameterization of their function approximators, ensuring that they hold for every parameter value. Our parameterization does not enforce them by construction, so we include their squared deviations in the training objective. We use the terminal loss $\mathcal{L}_T(\boldsymbol{\theta}) := \mathbb{E}[(V_{\boldsymbol{\theta}}(T, \boldsymbol{x}) - g(T, \boldsymbol{x}))^2]$ and the entropy-consistency loss

$$\mathcal{L}_C(\boldsymbol{\theta}, \boldsymbol{\varphi}) := \mathbb{E}\left[\left(\int_U \left[q_{\boldsymbol{\theta},\boldsymbol{\varphi}}(t, \boldsymbol{x}, \boldsymbol{u}) - \lambda \log \pi(\boldsymbol{u} \mid t, \boldsymbol{x})\right] \pi(\boldsymbol{u} \mid t, \boldsymbol{x})\, \mathrm{d}\boldsymbol{u}\right)^2\right],$$

where π is the current policy. The parameters are updated by

$$\begin{cases} \boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \eta_{\boldsymbol{\theta}} \left[\boldsymbol{S}_{\boldsymbol{\theta}} - w_T \nabla_{\boldsymbol{\theta}} \mathcal{L}_T(\boldsymbol{\theta})\right], \\ \boldsymbol{\varphi} \leftarrow \boldsymbol{\varphi} + \eta_{\boldsymbol{\varphi}} \left[\boldsymbol{S}_{\boldsymbol{\varphi}} - w_C \nabla_{\boldsymbol{\varphi}} \mathcal{L}_C(\boldsymbol{\theta}, \boldsymbol{\varphi})\right], \end{cases} \tag{38}$$

where $w_T, w_C \geq 0$ weight the terminal-condition and entropy-consistency losses, respectively. Algorithm 1 summarizes the resulting full-trajectory structured q-learning procedure.

After training, we use $V_{\boldsymbol{\theta}}$ to extract an approximate stopping region and free boundary for evaluating the original problem, and $q_{\boldsymbol{\theta},\boldsymbol{\varphi}}$ to determine investment in the continuation region. Policy randomization provides exploration during training, whereas both decisions are deterministic at execution. Following the treatment of deterministic stopping rules in Dai *et al.* (2026), we compare the approximate value function with the stopping payoff. With $w_{\boldsymbol{\theta}}(t, \boldsymbol{x}) := V_{\boldsymbol{\theta}}(t, \boldsymbol{x}) - g(t, \boldsymbol{x})$, define

$$\mathcal{C}_{\boldsymbol{\theta}} := \{(t, \boldsymbol{x}) : w_{\boldsymbol{\theta}}(t, \boldsymbol{x}) > 0\}, \qquad \mathcal{S}_{\boldsymbol{\theta}} := \{(t, \boldsymbol{x}) : w_{\boldsymbol{\theta}}(t, \boldsymbol{x}) \leq 0\},$$

where $\mathcal{C}_{\boldsymbol{\theta}}$ and $\mathcal{S}_{\boldsymbol{\theta}}$ denote the approximate continuation and stopping regions, whose boundary defines the approximate free boundary. On the discrete grid, the value-function approximation determines the stopping time $\tau_{\boldsymbol{\theta}} := \inf\{t_l : V_{\boldsymbol{\theta}}(t_l, \boldsymbol{X}_l) \leq g(t_l, \boldsymbol{X}_l)\} \wedge T$. Since the decision process terminates upon entering the stopping region, investment is specified in the continuation region. For $(t, \boldsymbol{x}) \in \mathcal{C}_{\boldsymbol{\theta}}$, the investment policy is selected according to $\boldsymbol{u}^{\alpha,*}(t, \boldsymbol{x}) \in \arg\max_{\boldsymbol{a} \in A} q_{\boldsymbol{\theta},\boldsymbol{\varphi}}\big(t, \boldsymbol{x}, (\boldsymbol{a}, 0)\big)$,

Algorithm 1: Full-trajectory structured q-learning

Input: initial state $\boldsymbol{x}_0$, horizon T, time step Δt, number of episodes N, number of time steps L, learning rates $\eta_{\boldsymbol{\theta}}, \eta_{\boldsymbol{\varphi}}$, penalty parameter K, temperature parameter λ, initial parameters $(\boldsymbol{\theta}_0, \boldsymbol{\varphi}_0)$, value-function approximator $V_{\boldsymbol{\theta}}$, and loss weights w_T, w_C;

Required program: an environment simulator $\boldsymbol{x}' = \text{Environment}_{\Delta t}(t, \boldsymbol{x}, \boldsymbol{u})$ that takes a time–state pair $(t, \boldsymbol{x})$ and an action $\boldsymbol{u} \in U$ and returns the next state $\boldsymbol{x}'$. The sampling policy is given by (34);

Initialize $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta}_0$ and $\boldsymbol{\varphi} \leftarrow \boldsymbol{\varphi}_0$;

for *episode* $n = 1, \ldots, N$ **do**

  Initialize $l = 0$, $\boldsymbol{X}_0 \leftarrow \boldsymbol{x}_0$, and $R_0 \leftarrow 1$;

  while $l < L$ **do**

    Sample $\boldsymbol{u}_l \sim \pi_{\boldsymbol{\theta},\boldsymbol{\varphi}}(\cdot \mid t_l, \boldsymbol{X}_l)$;

    Query the environment simulator: $\boldsymbol{x}' = \text{Environment}_{\Delta t}(t_l, \boldsymbol{X}_l, \boldsymbol{u}_l)$;

    Set $\boldsymbol{X}_{l+1} \leftarrow \boldsymbol{x}'$ using (35) and update R_{l+1} using (36);

    Set $l \leftarrow l + 1$;

  Compute $\Delta_l^{\boldsymbol{\theta},\boldsymbol{\varphi}}$ for $l = 0, \ldots, L-1$ using (37);

  Compute $\boldsymbol{S}_{\boldsymbol{\theta}}$, $\boldsymbol{S}_{\boldsymbol{\varphi}}$, $\mathcal{L}_T(\boldsymbol{\theta})$, and $\mathcal{L}_C(\boldsymbol{\theta}, \boldsymbol{\varphi})$;

  Update $\boldsymbol{\theta}$ and $\boldsymbol{\varphi}$ using (38);

Output: learned value function $V_{\boldsymbol{\theta}}$, structured q-function $q_{\boldsymbol{\theta},\boldsymbol{\varphi}}$, Gibbs policy $\pi_{\boldsymbol{\theta},\boldsymbol{\varphi}}$, and estimated market parameters $\boldsymbol{\varphi}$;

which maximizes the structured q-function over continuation actions. Since the stopping penalty vanishes when $u^\beta = 0$, this investment rule is equivalent to

$$\boldsymbol{u}^{\alpha,*}(t, \boldsymbol{x}) \in \arg\max_{\boldsymbol{a} \in A} \left\{ \mathcal{L}_{\boldsymbol{\varphi}}^{\boldsymbol{a}} V_{\boldsymbol{\theta}}(t, \boldsymbol{x}) + \mathcal{J}_{\boldsymbol{\varphi}}^{\boldsymbol{a}} V_{\boldsymbol{\theta}}(t, \boldsymbol{x}) \right\}. \tag{39}$$

We evaluate these deterministic rules in the original unregularized problem by applying $\boldsymbol{u}^{\alpha,*}$ and $\tau_{\boldsymbol{\theta}}$ in the test environment and estimating the initial value by Monte Carlo simulation:

$$\widehat{v}_{\text{orig}}^{\text{RL}}(0, \boldsymbol{x}_0) := \frac{1}{M} \sum_{m=1}^{M} \mathrm{e}^{-\rho \tau_{\boldsymbol{\theta},m}} g\left(\tau_{\boldsymbol{\theta},m}, \boldsymbol{X}^{*,m}_{\tau_{\boldsymbol{\theta},m}} \right), \tag{40}$$

where $\boldsymbol{X}^{*,m}$ is the state process on test path m under the feedback investment policy $\boldsymbol{u}^{\alpha,*}$, and $\tau_{\boldsymbol{\theta},m}$ is the stopping time prescribed by the above rule on that path. Test-path generation is described separately for the simulation and empirical studies. The estimator $\widehat{v}_{\text{orig}}^{\text{RL}}(0, \boldsymbol{x}_0)$ is used for comparison with the finite-difference benchmarks.

6. Numerical experiments

This section evaluates the structured q-learning method developed above using simulations and market data. In simulations, the true model parameters are known to the researcher but unobserved by the learning agent. We investigate the learned value function, investment policy, free boundary, and market parameters. The empirical analysis further evaluates the out-of-sample performance of the derived investment and stopping rules on asset data.

Table 1. True and initial values of market parameters

	μ	σ	ν	η
True value	0.1000	0.3000	0.2000	−0.0700
Initial guess	0.1400	0.3400	0.2600	−0.1000

6.1. *Simulation study*

We examine whether structured q-learning can jointly estimate the value function and market parameters from sample paths, without access to their true values, and derive investment and stopping decisions for a power-utility problem with jump-diffusion wealth dynamics. Our numerical benchmark is the finite-difference method, which uses the true market parameters to solve the penalty equation and approximate the original VI. The difference scheme is described in Appendix B.

6.1.1. **Problem setup and parameter selection.** Consider a market with a risk-free asset and a risky asset subject to jumps. The continuous-time portfolio choice framework and jump-diffusion asset model originate from Merton (1971) and Merton (1976), respectively. The risk-free asset satisfies $\mathrm{d}B_t = rB_t\,\mathrm{d}t$, and the risky asset price follows

$$\mathrm{d}S_t = (\mu - \nu J)S_{t-}\,\mathrm{d}t + \sigma S_{t-}\,\mathrm{d}W_t + JS_{t-}\,\mathrm{d}N_t, \quad J = \mathrm{e}^{\eta} - 1,$$

where W is a standard Brownian motion, N is a Poisson process with intensity ν, and J is the relative jump size. Writing $\alpha_t \in [0,1]$ for the fraction of wealth invested in the risky asset at time t, the wealth dynamics are

$$\mathrm{d}X_t = X_{t-}\left[r + \alpha_t(\mu - r - \nu J)\right]\mathrm{d}t + \alpha_t\sigma X_{t-}\,\mathrm{d}W_t + \alpha_t J X_{t-}\,\mathrm{d}N_t. \tag{41}$$

We use the power-utility stopping payoff $g(x) = (x - K_0)^{1-\gamma_1}/(1-\gamma_1)$ for $x \geq K_0$, with $g(K_0) = 0$, where K_0 is the absorbing exit wealth level. For an admissible investment policy α, let $\zeta^{\alpha}_{K_0} := \inf\{s \geq t : X^{\alpha}_s \leq K_0\}$ be the first time wealth reaches or falls below this level. Wealth follows (41) before $\zeta^{\alpha}_{K_0}$; on reaching K_0 or crossing below it after a negative jump, investment terminates immediately with zero payoff. Let $\mathcal{A}_t$ denote the admissible investment policies and $\mathcal{T}_{t,T}$ the stopping times taking values in $[t,T]$. The original investment and stopping problem has value function

$$v(t,x) = \sup_{\alpha\in\mathcal{A}_t,\,\tau\in\mathcal{T}_{t,T}} \mathbb{E}_{t,x}\left[\mathrm{e}^{-\rho(\tau-t)}g(X^{\alpha}_{\tau})\mathbf{1}_{\{\tau<\zeta^{\alpha}_{K_0}\}}\right], \quad (t,x)\in[0,T]\times[K_0,\infty), \tag{42}$$

with terminal condition $v(T,x) = g(x)$ and absorbing boundary condition $v(t,K_0) = 0$.

For the finite-difference benchmark, we use the true market parameters and the penalty equation to approximate the original VI, obtaining benchmark value functions, investment fractions and free boundaries. The penalty treatment and finite-difference scheme follow Forsyth and Vetzal (2002); see also Ma (2021) for discretization of the associated HJB equation. In contrast, the RL learner constructs the q-function without using the true market parameters and learns these parameters jointly with the value function from initial guesses. Training uses $L = 50$ time steps, batches of 64 sample paths and 2000 episodes. The investment fraction is restricted to $[0,1]$ and represented on a grid of 101 equally spaced points; integrals over this continuous action component in the Gibbs policy are evaluated by the trapezoidal rule. Table 1 reports the true and initial market parameters, and Table 2 gives the remaining model and training settings.

6.1.2. **Implementation details.** Since the structured q-function involves first- and second-order value-function derivatives, we represent $V_{\boldsymbol{\theta}}(t,x)$ by a feedforward neural network with smooth

Table 2. Main model parameters and training settings

Description	Symbol	Value	Description	Symbol	Value
Risk-free rate	r	0.05	Discount rate	ρ	0.06
Terminal time	T	1	Initial wealth	x_0	1.50
Absorbing exit wealth	K_0	1.00	Risk-aversion parameter	γ_1	0.50
Penalty parameter	K	10	Temperature parameter	λ	1
Number of time steps	L	50	Training episodes	N_{ep}	2000
Training batch size	M_{train}	64	Investment grid points	M_α	101
Value-function learning rate	$\eta_{\boldsymbol{\theta}}$	10^{-4}	Parameter learning rate	$\eta_{\boldsymbol{\varphi}}$	5×10^{-4}

activations. Its inputs are time, wealth surplus and numerically stabilized logarithmic features of wealth surplus. The network has three 128-unit hidden layers with Softplus activations and a scalar output; V_t, V_x and V_{xx} are obtained by automatic differentiation. Let $\boldsymbol{\varphi} := (\mu, \sigma, \nu, \eta)$ collect the unknown market parameters and set $J_{\boldsymbol{\varphi}} = \mathrm{e}^{\eta} - 1$. Under the structured parameterization in Section 5, the local diffusion and nonlocal jump operators are

$$\mathcal{L}^{\alpha}_{\boldsymbol{\varphi}} V_{\boldsymbol{\theta}}(t,x) = x\big[r + \alpha(\mu - r - \nu J_{\boldsymbol{\varphi}})\big] V_x(t,x) + \frac{1}{2}\alpha^2\sigma^2 x^2 V_{xx}(t,x),$$
$$\mathcal{J}^{\alpha}_{\boldsymbol{\varphi}} V_{\boldsymbol{\theta}}(t,x) = \nu\left[\widetilde{V}_{\boldsymbol{\theta}}\big(t, x(1+\alpha J_{\boldsymbol{\varphi}})\big) - V_{\boldsymbol{\theta}}(t,x)\right],$$

where $\widetilde{V}_{\boldsymbol{\theta}}$ denotes the zero extension at and below the absorbing boundary: $\widetilde{V}_{\boldsymbol{\theta}}(t,y) := V_{\boldsymbol{\theta}}(t,y)$ for $y > K_0$ and $\widetilde{V}_{\boldsymbol{\theta}}(t,y) := 0$ for $y \le K_0$. This extension accounts for negative jumps that reach or cross the absorbing boundary.

Define $F_{\boldsymbol{\theta},\boldsymbol{\varphi}}(t,x,\alpha) := \mathcal{L}^{\alpha}_{\boldsymbol{\varphi}} V_{\boldsymbol{\theta}}(t,x) + \mathcal{J}^{\alpha}_{\boldsymbol{\varphi}} V_{\boldsymbol{\theta}}(t,x)$. In implementation, $[0,1]$ is discretized as $A_h = \{a_1, \dots, a_{M_\alpha}\}$ with trapezoidal weights $\{\omega_i\}_{i=1}^{M_\alpha}$. The investment and stopping controls are sampled with probabilities

$$\mathbb{P}(\alpha = a_i \mid t,x) = \frac{\omega_i \exp(F_{\boldsymbol{\theta},\boldsymbol{\varphi}}(t,x,a_i)/\lambda)}{\displaystyle\sum_{j=1}^{M_\alpha} \omega_j \exp(F_{\boldsymbol{\theta},\boldsymbol{\varphi}}(t,x,a_j)/\lambda)}, \quad \mathbb{P}(\beta = 1 \mid t,x) = \frac{\exp(K[g(x) - V_{\boldsymbol{\theta}}(t,x)]/\lambda)}{1 + \exp(K[g(x) - V_{\boldsymbol{\theta}}(t,x)]/\lambda)},$$

where the additive dependence of $q_{\boldsymbol{\theta},\boldsymbol{\varphi}}$ on α and β allows the joint Gibbs policy to factorize into a Gibbs distribution over investment fractions and a Bernoulli distribution over the stopping control. Sequential sampling is therefore consistent with the joint policy, apart from the grid approximation of the continuous-action integral. Training uses the full-trajectory martingale orthogonality updates in Section 5, with the boundary loss $\mathcal{L}_B(\boldsymbol{\theta}) := \mathbb{E}[V_{\boldsymbol{\theta}}(t,K_0)^2]$ included to account for $v(t,K_0) = 0$.

6.1.3. **Convergence diagnostics.** Figure 1 shows the total training loss, an aggregate monitoring measure combining the martingale orthogonality, terminal-condition, absorbing-boundary and entropy-consistency losses with the implementation weights. It decreases rapidly early in training and then fluctuates at a relatively low level. This indicates that the updates reduce the martingale orthogonality residual and fitting errors, while training remains broadly stable.

Figure 2 plots the initial-state value estimate and relative error over training episodes. In panel (a), $V_{\boldsymbol{\theta}}(0,x_0)$ approximates the value function of the Shannon entropy-regularized exploration problem, changing substantially early in training and stabilizing later. The panel also shows the original-problem value estimate $\widehat{v}^{\text{RL}}_{\text{orig}}(0,x_0)$ from (40), which approaches the finite-difference benchmark $v_{\text{FDM}}(0,1.5) = 1.44986$ and remains close to it later in training. Define its relative error by

$$\text{RelErr}_v := \frac{\big|\widehat{v}^{\text{RL}}_{\text{orig}}(0,x_0) - v_{\text{FDM}}(0,x_0)\big|}{|v_{\text{FDM}}(0,x_0)|}, \tag{43}$$

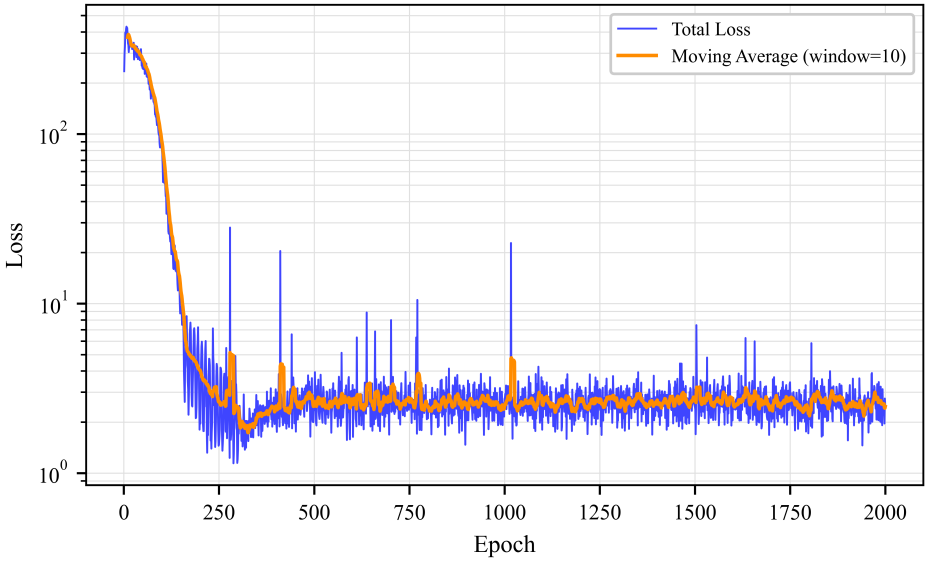


Figure 1. Training loss

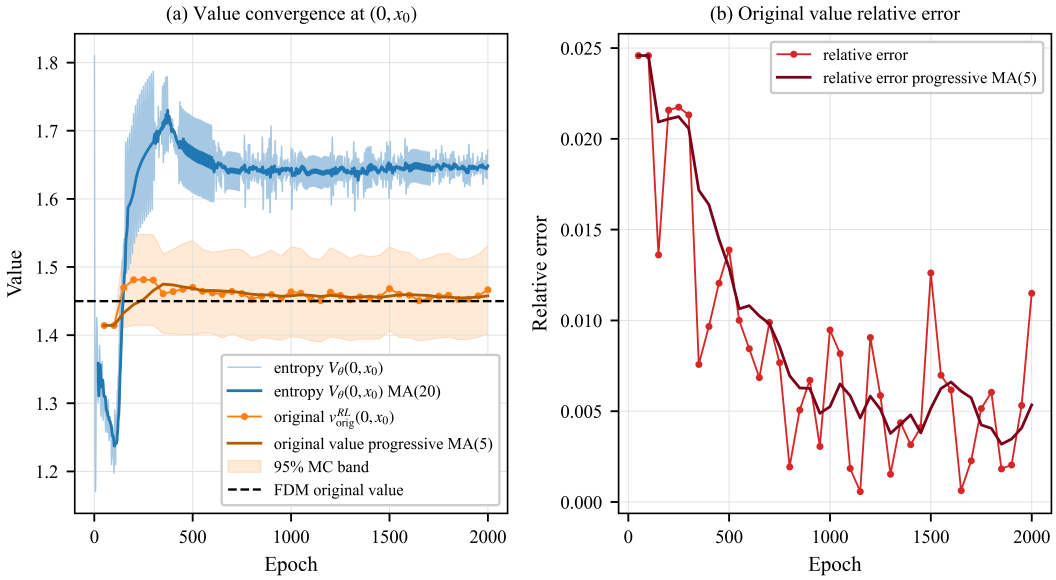


Figure 2. Value estimates and relative error

and panel (b) shows that this error generally decreases during training and remains low in later episodes. Local fluctuations reflect randomness in training paths, stochastic parameter updates and Monte Carlo evaluation. Re-evaluation after training on 1000 independent test paths gives $\widehat{v}^{\mathrm{RL}}_{\mathrm{orig}}(0, 1.5) = 1.46068$, with $\mathrm{RelErr}_v = 0.746\%$. Thus, both the exploration value approximation and the original-problem value estimate stabilize, and the extracted deterministic investment and stopping rules yield a value close to the finite-difference benchmark.

6.1.4. **Investment policy and free boundary.** Figure 3 compares the investment policies from structured q-learning and the finite-difference method. Panel (a) plots the learned investment fraction at $(0, x_0)$ over training episodes: it changes markedly early on, then fluctuates less and becomes relatively stable. Panel (b) shows pointwise relative errors over the (t, x) grid after training. Writing $\widehat{\alpha}_{\mathrm{RL}}(t_i, x_j)$ and $\alpha_{\mathrm{FDM}}(t_i, x_j)$ for the fractions produced by the two methods, respectively,

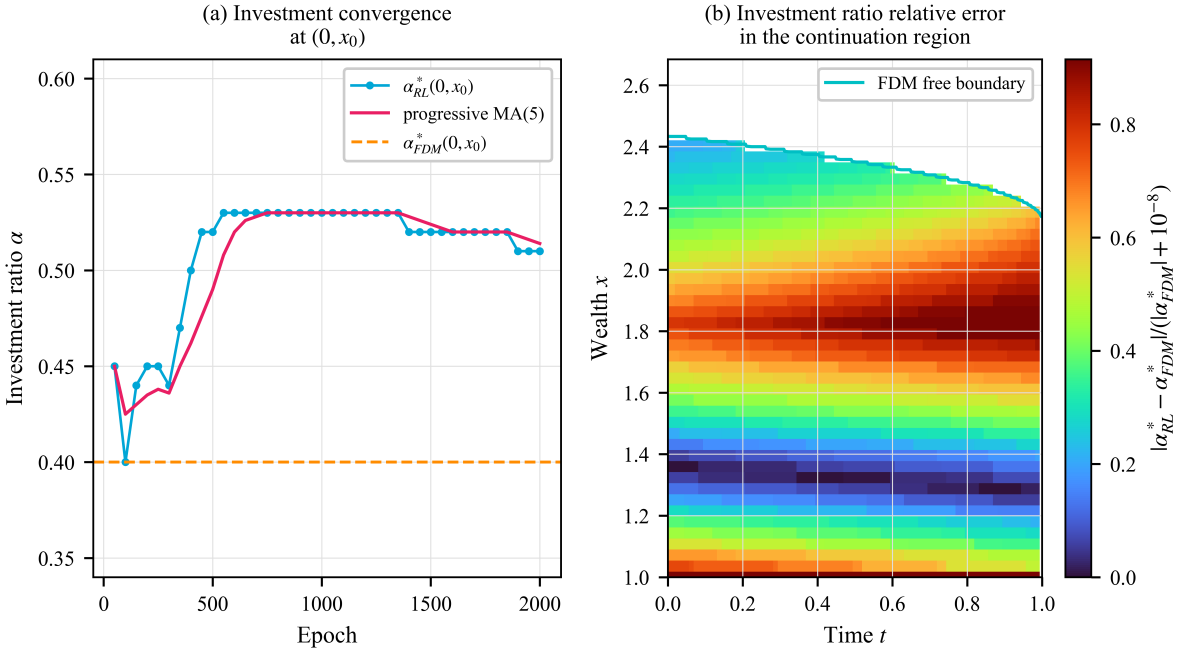


Figure 3. Investment policy convergence and relative error

we define their relative error by

$$\mathrm{RelErr}_\alpha(t_i, x_j) := \frac{|\widehat{\alpha}_{\mathrm{RL}}(t_i, x_j) - \alpha_{\mathrm{FDM}}(t_i, x_j)|}{|\alpha_{\mathrm{FDM}}(t_i, x_j)| + 10^{-8}},$$

where 10^{-8} prevents division by zero. The heatmap shows how errors vary across time and wealth. Differences are small at most grid points, indicating that the learned feedback investment policy is broadly consistent with the finite-difference benchmark.

Figure 4 compares the learned and finite-difference free boundaries. Red dots and green triangles mark states reached by test paths at discrete times and indicate the learned continue and stop decisions, respectively. The solid blue curve is the boundary extracted from $V_{\boldsymbol{\theta}}(t, x) - g(x)$, using a tolerance $\varepsilon_b = 10^{-4}$ to limit the effects of function approximation and grid discretization; the dashed black curve is the finite-difference boundary. The curves have similar profiles. To quantify boundary learning, we use the continuation-versus-stopping classification measure of Dai *et al.* (2026) on $M_{\mathrm{cls}} = 5000$ test paths. Let $X_l^{(m)}$ denote the state of path m at t_l and $b_{\mathrm{FDM}}(t_l)$ the finite-difference boundary. Define $c_{l,m}^{\mathrm{RL}} := \mathbf{1}_{\{V_{\boldsymbol{\theta}}(t_l, X_l^{(m)}) - g(X_l^{(m)}) \le 0\}}$ and $c_{l,m}^{\mathrm{FDM}} := \mathbf{1}_{\{X_l^{(m)} \ge b_{\mathrm{FDM}}(t_l)\}}$. Classification accuracy is $\mathrm{Acc}(t_l) := \frac{1}{M_{\mathrm{cls}}} \sum_{m=1}^{M_{\mathrm{cls}}} \mathbf{1}_{\{c_{l,m}^{\mathrm{RL}} = c_{l,m}^{\mathrm{FDM}}\}}$, for $l = 0, \ldots, L-1$. Average accuracy across time points exceeds 98%, with a minimum of approximately 95%. The learned rule therefore agrees closely with the finite-difference rule over the states reached by the test paths.

6.1.5. **Market parameter learning.** Figure 5 plots market-parameter estimates over training episodes. Whereas the finite-difference benchmark uses the true parameters, structured q-learning starts from the guesses in Table 1 and learns from sample paths. Each estimate moves towards its true value and becomes relatively stable later in training, although adjustment rates and final errors differ across parameters. The figure suggests that, in this simulation, the structured parameterization can use sample-path information to refine the initial market-parameter guesses.

To assess estimation accuracy, for $\vartheta \in \{\mu, \sigma, \nu, \eta\}$, we define the relative estimation error as $\mathrm{RelErr}(\widehat{\vartheta}) := |\widehat{\vartheta} - \vartheta_{\mathrm{true}}|/|\vartheta_{\mathrm{true}}|$. Table 3 reports final errors substantially smaller than those of the initial guesses, with all relative errors below 0.05. The volatility error is below 0.01, and the jump-

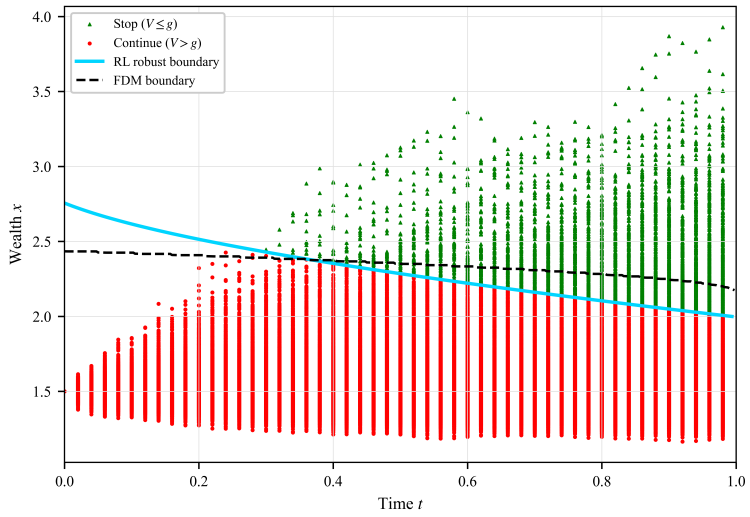


Figure 4. Stopping decisions and free-boundary comparison

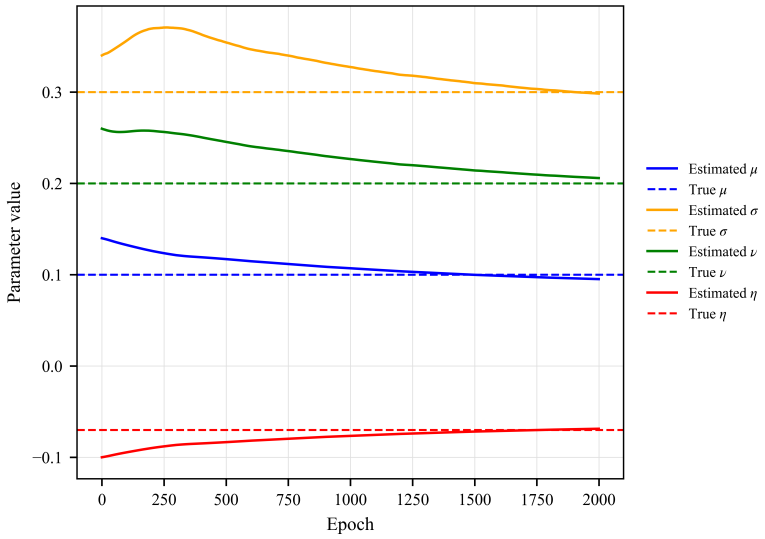


Figure 5. Market parameter estimates

Table 3. Market parameter learning results

Parameter	True value	Initial guess	Final estimate	Final relative error
μ	0.1000	0.1400	0.0952	0.0484
σ	0.3000	0.3400	0.2983	0.0055
ν	0.2000	0.2600	0.2058	0.0291
η	−0.0700	−0.1000	−0.0687	0.0181

intensity and jump-size errors are below 0.03. Under this simulation setup, the results indicate that the structured q-function can use diffusion and jump information in sample paths to estimate unknown market parameters while learning the value function and control policy.

6.1.6. **Sensitivity to the penalty and temperature parameters.** The penalty parameter K and temperature λ affect the stopping approximation and policy exploration, respectively. In the structured q-function, K weights the stopping-payoff gap $u^{\beta}[g(t,x) - V_{\boldsymbol{\theta}}(t,x)]$ and determines the decay rate of the auxiliary state when the stopping control is activated. The temperature controls the concentration of the Gibbs policy: larger λ gives a more dispersed action distribution,

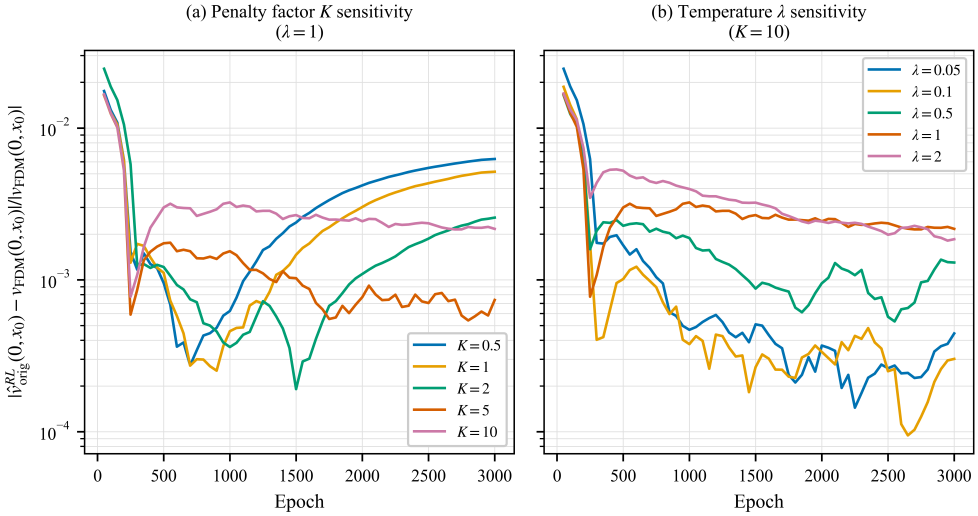


Figure 6. Sensitivity of relative value error to K and λ

Table 4. Parameter settings for the jump-risk sensitivity analysis

Risk scenario	Jump intensity ν	Jump-size parameter η	Relative jump size $J = \mathrm{e}^{\eta} - 1$
Low jump risk	0.10	-0.05	-0.0488
Moderate jump risk	1.50	-0.15	-0.1393
High jump risk	3.00	-0.25	-0.2212

while smaller λ places more mass on actions with larger q-values. We examine their effects on training using the relative original-problem value error, varying one parameter at a time as in Dai *et al.* (2026). For K, we fix $\lambda = 1$ and take $K \in \{0.5, 1, 2, 5, 10\}$; for λ, we fix $K = 10$ and take $\lambda \in \{0.05, 0.1, 0.5, 1, 2\}$. All other model and training settings are unchanged. Each configuration is trained for 3000 episodes; every 50 episodes, the original-problem value is estimated on 5000 independent test paths and its relative error computed using (43).

Figure 6(a) shows that errors for all penalty parameters fall sharply early in training and remain low thereafter. Increasing K from small to moderate values generally reduces late-stage error, suggesting a better approximation of the original stopping problem. Further increases beyond a certain level offer limited improvement, as time discretization, function approximation, stochastic optimization and Monte Carlo evaluation also contribute to the error. Choosing K therefore balances penalty-approximation accuracy against numerical training performance. Panel (b) shows rapid initial error reductions for all temperatures, with smaller λ generally yielding lower late-stage error. As λ decreases, the Gibbs policy concentrates on actions with larger q-values and approaches the deterministic execution rule; larger λ maintains a more dispersed distribution and greater exploration. These results illustrate the trade-off between exploration and execution-policy value accuracy. Errors decrease markedly with training across all configurations examined. The baseline sets $K = 10$ and $\lambda = 1$ to balance penalty approximation and exploration.

6.1.7. **Sensitivity to jump risk.** Unlike K and λ, jump intensity and size are market-risk parameters governing the frequency and magnitude of downward jumps, affecting continuation value, stopping and risky-asset allocation. We vary them to examine whether the learned decisions respond consistently with the model's economic mechanisms. With $J = \mathrm{e}^{\eta} - 1$, a lower $\eta < 0$ implies larger downward jumps, while higher ν implies more frequent jumps. Table 4 specifies low-, moderate- and high-risk scenarios. To isolate jump risk, all other market parameters are held fixed, with $K = 10$ and $\lambda = 1$ across scenarios.

Figures 7(a) and 7(b) show the entropy-regularized value function at $t = 0$ and its continuation

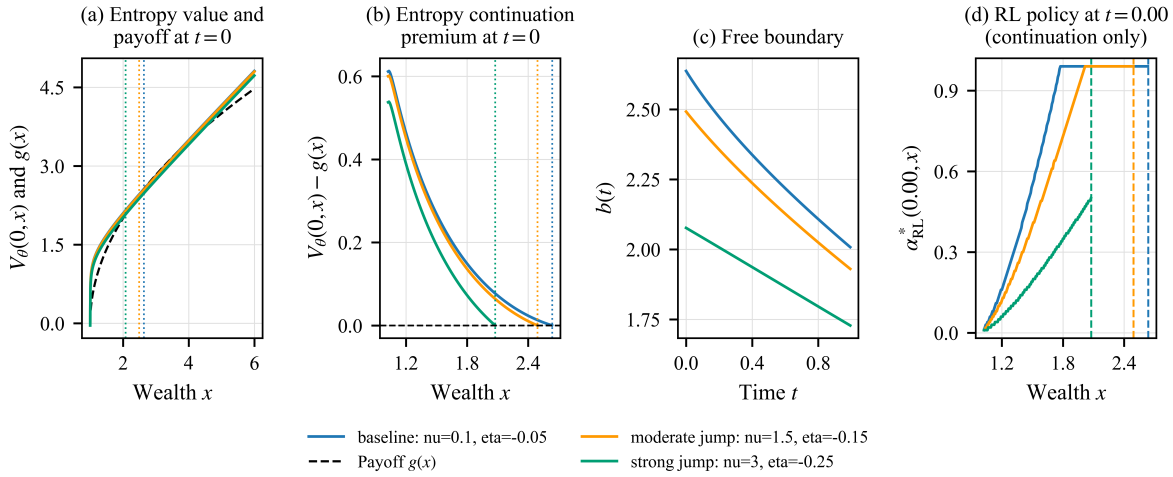


Figure 7. Value function, investment and stopping decisions under varying jump risk

premium over the stopping payoff, $V_{\boldsymbol{\theta}}(0,x)-g(x)$. As jump intensity and the magnitude of negative jumps increase, the premium generally declines and approaches zero at lower wealth levels. With expected-return parameters held fixed, greater downside jump risk reduces the value of continuing relative to immediate stopping, consistent with the risk aversion implied by the increasing, strictly concave power utility. Panel (c) shows a corresponding downward shift in the free boundary and contraction of the continuation region: the lower premium means that the stopping condition is met at a lower wealth level. The extracted boundary thus reflects the effect of downside jump risk on stopping. Panel (d) reports risky-asset investment fractions in the continuation region at $t = 0$. Under low jump risk, the fraction rises rapidly with wealth, whereas more frequent and larger negative jumps generally lead to lower exposure. At a given investment fraction, greater downside jump risk increases the likelihood of a large wealth loss, prompting a reduction in risky-asset holdings. Overall, greater downside jump risk reduces the value advantage of continued investment, so the stopping condition is met at a lower wealth level and risky-asset holdings decline. The stopping rule and investment fraction respond coherently to jump risk, consistent with the model's economic intuition, verifying the rationality of decisions yielded by structured q-learning.

6.2. *Empirical study*

We evaluate the out-of-sample performance of the investment and stopping rules obtained from structured q-learning using real-market data, with MLE–FDM as the benchmark. We describe the data and experimental setup, the bootstrap evaluation procedure and the results.

6.2.1. **Data and experimental setup.** The empirical study considers the Huatai-PineBridge CSI 300 ETF (510300), a broad-based Chinese ETF tracking the CSI 300 Index, and the ARK Innovation ETF (ARKK), an actively managed US innovation-themed ETF. The two funds differ substantially in their underlying markets and product characteristics. Daily prices for 510300 and ARKK are obtained from Eastmoney and Yahoo Finance, respectively, and daily log returns are computed from adjusted closing prices. For each asset, the sample is divided chronologically into training and test periods. Training-period returns are used for both MLE–FDM parameter estimation and RL training, whereas test-period returns are withheld from both procedures and used only for out-of-sample evaluation. Table 5 reports the sample periods, return counts and risk-free rates. For 510300, we use Chinese one-year government bond yields from ChinaBond; for ARKK, we use US one-year constant-maturity Treasury yields from FRED. For each asset, the fixed annualized risk-free rate is the arithmetic mean of the yields matched to its training-period trading

Table 5. Empirical sample periods and risk-free rates

Description	510300	ARKK
Training period	2012-05-29–2019-12-31	2014-11-03–2021-12-31
Test period	2020-01-02–2025-12-31	2022-01-03–2025-12-31
Training return observations	1851	1804
Test return observations	1455	1003
Annualized risk-free rate r	0.029069	0.009797

Table 6. Market parameters from MLE and structured q-learning

Asset	Source	μ	σ	ν	η
510300	MLE estimate	0.1006	0.1998	4.8008	−0.0529
510300	RL initial guess	0.1500	0.2200	5.0000	−0.0750
510300	RL final estimate	0.1063	0.1988	4.7158	−0.0528
ARKK	MLE estimate	0.2746	0.2693	7.3204	−0.0548
ARKK	RL initial guess	0.3900	0.2800	7.5000	−0.0800
ARKK	RL final estimate	0.2768	0.2722	7.3557	−0.0545

dates. Both assets have a one-year investment horizon, discretized into 252 trading days. Wealth is measured relative to the absorbing exit level $K_0 = 1$, with initial wealth $x_0 = 1.5$, to make the utility payoffs and stopping rules comparable.

Training paths for structured q-learning are generated from training-period returns using the nonparametric bootstrap of Efron (1979). For each one-year path, we independently sample 252 daily log returns with replacement from their empirical distribution. Let Y_l denote the sampled log return for interval l. Before absorption, wealth evolves as $X_{l+1} = X_l\left[(1-\alpha_l)\mathrm{e}^{r\Delta t} + \alpha_l \mathrm{e}^{Y_l}\right]$, with paths terminated upon reaching K_0, as in the simulation study. The remaining training and policy-extraction steps follow the structured q-learning algorithm in Section 5. For the MLE–FDM benchmark, we fit the fixed-jump-size jump-diffusion model to training-period daily log returns and use the estimated parameters in the finite-difference solver to obtain the value function, investment policy and free boundary. Maximum likelihood estimation uses a Poisson–Gaussian mixture likelihood; see Ball and Torous (1983) for the corresponding distributional representation. In contrast, structured q-learning starts from specified parameter guesses and jointly updates the value-function weights and market parameters using the bootstrapped training paths. Table 6 reports the parameter results for both methods.

Table 6 shows that the structured q-learning estimates move from their initial guesses toward the corresponding MLE estimates and end relatively close to them. Although MLE and structured q-learning use different parameter-learning criteria, their estimates are broadly comparable in magnitude. In this experiment, joint learning from sample paths therefore substantially revises the initial parameter guesses.

6.2.2. **Out-of-sample bootstrap evaluation.** Out-of-sample evaluation uses the empirical distribution of test-period daily log returns. Following an evaluation design similar to Gao *et al.* (2026), we use a nonparametric bootstrap to generate $M = 5000$ one-year test return sequences for each asset, with 252 daily returns per sequence. On each common sequence, structured q-learning and MLE–FDM start from $x_0 = 1.5$ and apply their respective investment and stopping rules. Let τ_m denote the stopping time selected by a strategy on sequence m, and let $\zeta^{\Delta t}_{K_0,m}$ denote its discrete absorbing-boundary hitting time. The effective termination time is $\widehat{\tau}_m := \tau_m \wedge \zeta^{\Delta t}_{K_0,m}$, and the realized objective is $U_m := \mathrm{e}^{-\rho\widehat{\tau}_m} g(X^{(m)}_{\widehat{\tau}_m})$, with $U_m = 0$ upon absorption. In this subsection, MU and std_U denote the sample mean and sample standard deviation of the realized objective across test paths, defined by $\mathrm{MU} := \frac{1}{M}\sum_{m=1}^{M} U_m$, $\mathrm{std}_U := \left[\frac{1}{M-1}\sum_{m=1}^{M}\big(U_m - \mathrm{MU}\big)^2\right]^{1/2}$. We also report the mean effective termination time $\overline{\tau} := M^{-1}\sum_{m=1}^{M}\widehat{\tau}_m$ and the mean and dispersion of

Table 7. Out-of-sample comparison of structured q-learning and MLE–FDM

Asset	Method	MU	std_U	$\overline{\tau}$	MR	std_R
510300	Structured q-learning	1.4083	0.1646	0.2064	0.0322	0.0669
510300	MLE–FDM	1.3859	0.3012	1.0000	0.0447	0.1647
ARKK	Structured q-learning	1.2919	0.6410	0.4848	0.0297	0.2312
ARKK	MLE–FDM	1.1419	0.9124	0.9963	0.0671	0.5172

wealth returns at the one-year horizon. Let $\widetilde{X}_{T,m}$ denote the wealth used to calculate the horizon-end return on path m. If a strategy stops voluntarily before T, its stopping wealth is carried forward at the risk-free rate, so that $\widetilde{X}_{T,m} = X^{(m)}_{\widehat{\tau}_m} \mathrm{e}^{r(T-\widehat{\tau}_m)}$. If the path reaches the absorbing boundary, wealth is fixed at K_0, giving $\widetilde{X}_{T,m} = K_0$; if the strategy continues until T, then $\widetilde{X}_{T,m} = X^{(m)}_T$. With the pathwise wealth return defined as $R_m := (\widetilde{X}_{T,m} - x_0)/x_0$, we define its cross-path sample mean MR and sample standard deviation std_R at the one-year horizon by $\mathrm{MR} = \frac{1}{M}\sum_{m=1}^{M} R_m$, $\mathrm{std}_R = \left[\frac{1}{M-1}\sum_{m=1}^{M}(R_m - \mathrm{MR})^2\right]^{1/2}$.

Table 7 compares structured q-learning and MLE–FDM on the same 5000 bootstrap test paths. Structured q-learning yields a higher mean objective value for both assets. For 510300, MU is approximately 1.61% higher than under MLE–FDM, while std_U and std_R are about 45.37% and 59.37% lower, respectively. For ARKK, MU is about 13.13% higher, while std_U and std_R are approximately 29.74% and 55.30% lower. Structured q-learning also terminates earlier on average and produces a lower mean wealth return for both assets. Earlier stopping shortens exposure to the risky asset, consistent with the lower dispersion of horizon-end wealth returns. In the objective $\mathrm{e}^{-\rho\tau} g(X_\tau)$, stopping time affects discounting and stopping wealth enters through utility; together, they determine the payoff. Thus, the lower mean wealth return coexists with a higher MU and a lower std_U for both assets. Under the sample split and bootstrap evaluation used here, the learned investment and stopping rules perform favourably on the original problem's objective.

7. Conclusion

We have developed a structured continuous-time q-learning method for investment and stopping in a jump-diffusion market with unknown parameters. Drawing on the penalty-based mixed-control formulation of Dai *et al.* (2026), the continuous-time q-learning framework of Jia and Zhou (2023), its jump-diffusion treatment in Bo *et al.* (2026) and the martingale policy-evaluation approach of Jia and Zhou (2022a), we develop a sample-path-based learning procedure for investment and stopping decisions using a structured q-function that retains the controlled generator and stopping-penalty terms, enabling joint updates of the value-function parameters and unknown market parameters. In simulations, the learned rules yield original-problem value estimates, investment policies and free boundaries close to the finite-difference benchmark, while market-parameter estimates move from their initial guesses towards the true values and stabilize nearby. For the two assets considered, out-of-sample bootstrap evaluation shows higher mean objective values and lower objective and wealth-return dispersion than MLE–FDM. These results indicate that incorporating model structure into continuous-time q-learning provides a viable sample-path-based method for solving investment and stopping problems with jump risk and unknown market parameters. The method still relies on a specified jump-diffusion generator. Future work could establish convergence and error bounds and extend the framework to more general mixed-control settings with random jump sizes, state-dependent jump intensities and multiple risky assets.

Declaration of generative AI. The authors employed GPT-6.1 for language polishing. All AI revisions were reviewed by the authors, who verified the originality and accuracy of the content and references. The authors take full responsibility for the final manuscript.

Data availability. Our manuscript has no associated data.

Acknowledgement. T. Zhang is supported by the National Natural Science of Foundation of China (No.12401619) and the Natural Science Foundation of the Jiangsu Higher Education Institutions of China (No. 24KJB110023).

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Appendix A: Proofs of the main results

A.1. *Proof of Theorem 2.3*

To apply the comparison and penalty convergence results of Reisinger and Zhang (2021), we show that, under Assumptions 2.1 and 2.2, the HJB VI (3) and the penalized equation (5) become special cases of their equations (1.1) and (3.1), respectively, after reversing time. Set $U(s,\boldsymbol{x}) = v(T-s,\boldsymbol{x})$, $U^K(s,\boldsymbol{x}) = v^K(T-s,\boldsymbol{x})$, $\zeta(s,\boldsymbol{x}) = g(T-s,\boldsymbol{x})$ and $g_0(\boldsymbol{x}) = g(T,\boldsymbol{x})$, and write

$$\widehat{\boldsymbol{b}}^{\alpha}(t,\boldsymbol{x}) := \boldsymbol{b}(t,\boldsymbol{x},\boldsymbol{\alpha}) + \int_{\mathbb{R}} \boldsymbol{\gamma}(t,\boldsymbol{x},\boldsymbol{\alpha},z)\, m(\mathrm{d}z).$$

In their equations, take the drift, diffusion coefficient and jump amplitude to be $\widehat{\boldsymbol{b}}^{\alpha}(T-s,\boldsymbol{x})$, $\boldsymbol{\sigma}(T-s,\boldsymbol{x},\boldsymbol{\alpha})$, and $\boldsymbol{\gamma}(T-s,\boldsymbol{x},\boldsymbol{\alpha},z)$, respectively, and take the penalty parameter to be K and the driver to be $f^{\alpha}(s,\boldsymbol{x},y,p,k) = -\rho y$. Set their additional nonlocal weight and nonlinear map to zero. The drift correction cancels the gradient compensation in the jump operator, leaving the total generator unchanged. Assumption 2.2 gives $\boldsymbol{\gamma}(t,\boldsymbol{x},\boldsymbol{\alpha},0) = 0$, so restricting m to $E = \mathbb{R}\setminus\{0\}$ leaves the jump operator unchanged. With $\nu := m|_E$, $\int_E (1\wedge|z|^2)\,\nu(\mathrm{d}z) \le \nu(E) \le m(\mathbb{R}) < \infty$. Thus the jump measure satisfies the required condition. Moreover,

$$\|\widehat{\boldsymbol{b}}^{\alpha}\|_1 \le \|\boldsymbol{b}(\cdot,\cdot,\boldsymbol{\alpha})\|_1 + \int_{\mathbb{R}} \|\boldsymbol{\gamma}(\cdot,\cdot,\boldsymbol{\alpha},z)\|_1\, m(\mathrm{d}z) \le C + C\int_{\mathbb{R}} (1\wedge|z|)\, m(\mathrm{d}z) \le C\big(1+m(\mathbb{R})\big),$$

so $\widehat{\boldsymbol{b}}^{\alpha}$ is bounded and spatially Lipschitz, uniformly in $\boldsymbol{\alpha}$. The continuity assumptions on $\boldsymbol{b}$ and $\boldsymbol{\gamma}$, together with dominated convergence, also give continuity of $\widehat{\boldsymbol{b}}^{\alpha}(t,\boldsymbol{x})$ in $(t,\boldsymbol{\alpha})$ for each fixed $\boldsymbol{x}$. The driver has monotonicity constant $-\rho$ and growth function $\varphi(q) = \rho q$, and is independent of its gradient and additional nonlocal inputs. Finally, $g_0(\boldsymbol{x}) = \zeta(0,\boldsymbol{x})$, so the initial compatibility condition holds with equality.

Proposition A.1 *Fix the market coefficients, jump measure and discount rate. Let g_1 and g_2 be two stopping payoffs such that Assumptions 2.1 and 2.2 hold in each case. Let w_1 and w_2 be a bounded viscosity subsolution and supersolution, respectively, of the original HJB VI with stopping payoffs g_1 and g_2. For fixed $K \ge 0$, let w_1^K and w_2^K be a bounded viscosity subsolution and supersolution, respectively, of the corresponding penalized equations. Then, for every $(t,\boldsymbol{x}) \in [0,T]\times\mathbb{R}^n$,*

$$\begin{aligned} w_1(t,\boldsymbol{x}) - w_2(t,\boldsymbol{x}) &\le \big|(w_1(T,\cdot) - w_2(T,\cdot))^+\big|_0 + |g_1 - g_2|_0, \\ w_1^K(t,\boldsymbol{x}) - w_2^K(t,\boldsymbol{x}) &\le \big|(w_1^K(T,\cdot) - w_2^K(T,\cdot))^+\big|_0 + |g_1 - g_2|_0. \end{aligned} \tag{A1}$$

where $|h|_0 := \sup_{(t,\boldsymbol{x})\in[0,T]\times\mathbb{R}^n} |h(t,\boldsymbol{x})|$ and $|h(T,\cdot)|_0 := \sup_{\boldsymbol{x}\in\mathbb{R}^n} |h(T,\boldsymbol{x})|$.

Proof. Under the preceding time change and generator identification, apply Reisinger and Zhang (2021, Proposition 3.1) with obstacles $\zeta_i(s,\boldsymbol{x}) = g_i(T-s,\boldsymbol{x})$, $i = 1,2$, and identical remaining coefficients. Setting $s = T - t$ in the resulting estimates yields the two inequalities. □

Corollary A.2 *Suppose that Assumptions 2.1 and 2.2 hold. The following comparison statements hold.*

- *(i) Let w_1 and w_2 be a bounded viscosity subsolution and supersolution, respectively, of the same HJB VI (3). If $w_1(T,\boldsymbol{x}) \le w_2(T,\boldsymbol{x})$ for every $\boldsymbol{x}\in\mathbb{R}^n$, then $w_1(t,\boldsymbol{x}) \le w_2(t,\boldsymbol{x})$ for all $(t,\boldsymbol{x}) \in [0,T]\times\mathbb{R}^n$.*
- *(ii) For each fixed $K \ge 0$, the same conclusion holds when w_1 and w_2 are a bounded viscosity subsolution and supersolution, respectively, of the same penalized equation (5).*

Proof. For (i), the two functions are a subsolution and a supersolution of the same HJB VI, so $g_1 = g_2 = g$ in Proposition A.1. Its first estimate therefore gives

$$w_1(t,\boldsymbol{x}) - w_2(t,\boldsymbol{x}) \le \big|(w_1(T,\cdot) - w_2(T,\cdot))^+\big|_0 = 0.$$

The equality follows from the assumed terminal ordering, which proves (i). For (ii), use the second estimate in that proposition with the same stopping payoff and the same penalty parameter for both functions. The stopping-payoff difference again vanishes, and the terminal ordering gives the same inequality. □

Remark 1 Let v_1 and v_2 be two bounded continuous viscosity solutions of the same HJB VI (3), or of the same penalized equation (5) with the same parameter K. Each solution is both a subsolution and a supersolution, and $v_1(T,\boldsymbol{x}) = v_2(T,\boldsymbol{x}) = g(T,\boldsymbol{x})$. Applying Corollary A.2 first with v_1 as the subsolution and v_2 as the supersolution gives $v_1 \le v_2$. Reversing their roles gives $v_2 \le v_1$. Thus $v_1 = v_2$, establishing uniqueness in this class.

THEOREM A.3 *Suppose that Assumptions 2.1 and 2.2 hold. Let v and v^K be the bounded viscosity solutions of* (3) *and* (5)*, respectively, with $K \ge 0$. Then, for any $0 \le K_1 \le K_2$,*

$$v^{K_1}(t,\boldsymbol{x}) \le v^{K_2}(t,\boldsymbol{x}) \le v(t,\boldsymbol{x}), \qquad (t,\boldsymbol{x}) \in [0,T]\times\mathbb{R}^n.$$

Proof. Since v^{K_2} is a viscosity solution of (5), it is both a subsolution and a supersolution. We use these two properties separately. For a smooth test function ϕ, write

$$\mathcal{A}\phi := -\phi_t + \rho\phi - \sup_{\boldsymbol{\alpha}\in A}\big(\mathcal{L}^{\alpha}\phi + \mathcal{J}^{\alpha}\phi\big).$$

First, take any $(t_0,x_0) \in [0,T)\times\mathbb{R}^n$ and $\phi \in C^{1,2}([0,T]\times\mathbb{R}^n)$ such that $\phi(t_0,x_0) = v^{K_2}(t_0,x_0)$ and $v^{K_2} - \phi$ attains its global maximum there. The subsolution property implies

$$\mathcal{A}\phi(t_0,x_0) - K_2\big(g(t_0,x_0) - v^{K_2}(t_0,x_0)\big)^+ \le 0.$$

If $v^{K_2}(t_0,x_0) \le g(t_0,x_0)$, the obstacle branch is nonpositive. If $v^{K_2}(t_0,x_0) > g(t_0,x_0)$, the penalty vanishes, so $\mathcal{A}\phi(t_0,x_0) \le 0$. In either case,

$$\min\big\{v^{K_2}(t_0,x_0) - g(t_0,x_0),\, \mathcal{A}\phi(t_0,x_0)\big\} \le 0.$$

Thus v^{K_2} satisfies the viscosity subsolution condition for the original HJB VI. It also satisfies the prescribed terminal condition. Since v is a viscosity solution of the HJB VI, it is in particular a supersolution of that equation. The equality $v^{K_2}(T,\boldsymbol{x}) = v(T,\boldsymbol{x})$ and Corollary A.2(i) give $v^{K_2} \le v$.

Next, take any $(t_0,x_0) \in [0,T)\times\mathbb{R}^n$ and $\phi \in C^{1,2}([0,T]\times\mathbb{R}^n)$ such that $\phi(t_0,x_0) = v^{K_2}(t_0,x_0)$ and $v^{K_2} - \phi$ attains its global minimum there. The supersolution property gives

$$\mathcal{A}\phi(t_0,x_0) - K_2\big(g(t_0,x_0) - v^{K_2}(t_0,x_0)\big)^+ \ge 0.$$

Since $K_2 - K_1 \ge 0$, it follows that

$$\begin{aligned}&\mathcal{A}\phi(t_0,x_0) - K_1\big(g(t_0,x_0) - v^{K_2}(t_0,x_0)\big)^+\\ &= \mathcal{A}\phi(t_0,x_0) - K_2\big(g(t_0,x_0) - v^{K_2}(t_0,x_0)\big)^+ + (K_2-K_1)\big(g(t_0,x_0) - v^{K_2}(t_0,x_0)\big)^+ \ge 0.\end{aligned}$$

Hence v^{K_2} satisfies the viscosity supersolution condition for the penalized equation with parameter K_1, together with its prescribed terminal condition. The function v^{K_1}, being a viscosity solution of

that equation, is also a subsolution. Since $v^{K_1}(T,\boldsymbol{x}) = v^{K_2}(T,\boldsymbol{x})$, Corollary A.2(ii), applied with the fixed parameter K_1, gives $v^{K_1} \le v^{K_2}$. Combining the two comparisons proves the result. □

Proof of Theorem 2.3. By the preceding time change and generator identification, Reisinger and Zhang (2021, Theorem 3.2) gives $U^K(s,\boldsymbol{x}) \uparrow U(s,\boldsymbol{x})$ as $K \to \infty$ for every $(s,\boldsymbol{x}) \in [0,T]\times\mathbb{R}^n$. Since $U^K(s,\boldsymbol{x}) = v^K(T-s,\boldsymbol{x})$ and $U(s,\boldsymbol{x}) = v(T-s,\boldsymbol{x})$, setting $s = T-t$ yields $v^K(t,\boldsymbol{x}) \uparrow v(t,\boldsymbol{x})$ for every $(t,\boldsymbol{x}) \in [0,T]\times\mathbb{R}^n$. □

A.2. *Proof of Theorem 2.4*

Proof. Fix an initial time–state pair $(s,\boldsymbol{x}) \in [0,T)\times\mathbb{R}^n$ and an admissible control $\boldsymbol{u} \in \mathcal{U}_s$. Write $\boldsymbol{X} = \boldsymbol{X}^{\boldsymbol{u}}$ for the corresponding state process and set $\Lambda_t := \exp\{-\rho(t-s) - \int_s^t f(r,\boldsymbol{X}_r,\boldsymbol{u}_r)\,\mathrm{d}r\}$. For a sufficiently regular function V, define

$$h_V(t,\boldsymbol{x},\boldsymbol{u}) := V_t - \rho V + \mathcal{L}^{\boldsymbol{u}^\alpha} V + \mathcal{J}^{\boldsymbol{u}^\alpha} V - fV + G,$$

where all terms on the right-hand side are evaluated at $(t,\boldsymbol{x},\boldsymbol{u})$. Equation (11) implies $h_V(t,\boldsymbol{x},\boldsymbol{u}) \le 0$ for every $\boldsymbol{u} \in U$. Applying the jump-diffusion Itô formula to $V(t,\boldsymbol{X}_t)$ and the product rule to the discounted process $\Lambda_t V(t,\boldsymbol{X}_t)$ yields

$$\Lambda_t V(t,\boldsymbol{X}_t) + \int_s^t \Lambda_r G(r,\boldsymbol{X}_r,\boldsymbol{u}_r)\,\mathrm{d}r = V(s,\boldsymbol{x}) + \int_s^t \Lambda_r h_V(r,\boldsymbol{X}_{r-},\boldsymbol{u}_r)\,\mathrm{d}r + M_t^{\boldsymbol{u}}, \tag{A2}$$

where the local martingale $M^{\boldsymbol{u}}$ is

$$\begin{aligned} M_t^{\boldsymbol{u}} = &\int_s^t \Lambda_r V_{\boldsymbol{x}}(r,\boldsymbol{X}_{r-})^\top \boldsymbol{\sigma}(r,\boldsymbol{X}_{r-},\boldsymbol{u}_r^\alpha)\,\mathrm{d}\boldsymbol{W}_r \\ &+ \int_s^t \int_{\mathbb{R}} \Lambda_r \big[V(r,\boldsymbol{X}_{r-} + \boldsymbol{\gamma}(r,\boldsymbol{X}_{r-},\boldsymbol{u}_r^\alpha,z)) - V(r,\boldsymbol{X}_{r-})\big]\big(N(\mathrm{d}r,\mathrm{d}z) - m(\mathrm{d}z)\,\mathrm{d}r\big). \end{aligned}$$

For $r < T$, define

$$\begin{aligned} Q_r^{\boldsymbol{u}} := &\int_s^r \Lambda_\ell^2 \big|V_{\boldsymbol{x}}(\ell,\boldsymbol{X}_{\ell-})^\top \boldsymbol{\sigma}(\ell,\boldsymbol{X}_{\ell-},\boldsymbol{u}_\ell^\alpha)\big|^2\,\mathrm{d}\ell \\ &+ \int_s^r \int_{\mathbb{R}} \Lambda_\ell \big|V(\ell,\boldsymbol{X}_{\ell-} + \boldsymbol{\gamma}(\ell,\boldsymbol{X}_{\ell-},\boldsymbol{u}_\ell^\alpha,z)) - V(\ell,\boldsymbol{X}_{\ell-})\big|\,m(\mathrm{d}z)\,\mathrm{d}\ell, \end{aligned}$$

and let $\tau_n := \inf\{r \in [s,T) : Q_r^{\boldsymbol{u}} \ge n\} \wedge T$, with $\tau_n = T$ if the threshold is not reached. The regularity conditions in the theorem and Assumption 2.1 imply that, for fixed $t < T$, $t\wedge\tau_n \uparrow t$ and $\mathbb{E}_{s,\boldsymbol{x}}[M_{t\wedge\tau_n}^{\boldsymbol{u}}] = 0$. Apply (A2) at $t\wedge\tau_n$, take expectations and use $h_V \le 0$:

$$V(s,\boldsymbol{x}) \ge \mathbb{E}_{s,\boldsymbol{x}}\left[\Lambda_{t\wedge\tau_n} V(t\wedge\tau_n,\boldsymbol{X}_{t\wedge\tau_n}) + \int_s^{t\wedge\tau_n} \Lambda_r G(r,\boldsymbol{X}_r,\boldsymbol{u}_r)\,\mathrm{d}r\right]. \tag{A3}$$

Since $f \ge -c$ and V and G satisfy the stated growth bounds, there is a constant $C > 0$, independent of n and t, such that

$$\left|\Lambda_{t\wedge\tau_n} V(t\wedge\tau_n,\boldsymbol{X}_{t\wedge\tau_n}) + \int_s^{t\wedge\tau_n} \Lambda_r G(r,\boldsymbol{X}_r,\boldsymbol{u}_r)\,\mathrm{d}r\right| \le C\left(1 + \sup_{r\in[s,T]} |\boldsymbol{X}_r|^p\right),$$

where the right-hand side is integrable by Assumption 2.1. Letting first $n \to \infty$ and then $t \uparrow T$, dominated convergence and the terminal condition yield

$$V(s,\boldsymbol{x}) \geq \mathbb{E}_{s,\boldsymbol{x}}\left[\int_s^T \Lambda_r G(r,\boldsymbol{X}_r,\boldsymbol{u}_r)\,\mathrm{d}r + \Lambda_T H(T,\boldsymbol{X}_T)\right] = \bar{J}(s,\boldsymbol{x};\boldsymbol{u}).$$

As $\boldsymbol{u}$ was arbitrary, $\mathcal{V}(s,\boldsymbol{x}) \leq V(s,\boldsymbol{x})$.

If the feedback control $\boldsymbol{u}^*$ in the theorem exists and is admissible, its pointwise maximization property gives $h_V(r,\boldsymbol{X}_{r-}^{\boldsymbol{u}^*},\boldsymbol{u}_r^*) = 0$ along the corresponding state process. Apply (A2) at $t\wedge\tau_n$ and take expectations, now constructing Λ and τ_n under $\boldsymbol{u}^*$:

$$V(s,\boldsymbol{x}) = \mathbb{E}_{s,\boldsymbol{x}}\left[\Lambda_{t\wedge\tau_n} V(t\wedge\tau_n,\boldsymbol{X}_{t\wedge\tau_n}^{\boldsymbol{u}^*}) + \int_s^{t\wedge\tau_n} \Lambda_r G(r,\boldsymbol{X}_r^{\boldsymbol{u}^*},\boldsymbol{u}_r^*)\,\mathrm{d}r\right].$$

Passing first to $n\to\infty$ and then to $t\uparrow T$ as above gives $V(s,\boldsymbol{x}) = \bar{J}(s,\boldsymbol{x};\boldsymbol{u}^*)$. Consequently, $V(s,\boldsymbol{x}) = \bar{J}(s,\boldsymbol{x};\boldsymbol{u}^*) \leq \mathcal{V}(s,\boldsymbol{x}) \leq V(s,\boldsymbol{x})$, so $V=\mathcal{V}$ and $\boldsymbol{u}^*$ is an optimal feedback control for (10). The assertion at $s=T$ follows directly from the terminal condition. □

A.3. *Proof of Theorem 3.2*

Proof. Fix $(t,\boldsymbol{x}) \in [0,T)\times\mathbb{R}^n$ and $\pi \in \Pi_t$, and write $\pi' := \mathcal{I}(\pi) \in \Pi_t$. We evaluate J^π along the relaxed state process starting from $\bar{\boldsymbol{X}}_t^{\pi'} = \boldsymbol{x}$. Define $\bar{\beta}^{\pi'}(s,\boldsymbol{y}) := \int_U u^\beta \pi'(\boldsymbol{u}\mid s,\boldsymbol{y})\,\mathrm{d}\boldsymbol{u}$ and $\Lambda_s := \exp\{-\rho(s-t) - K\int_t^s \bar{\beta}^{\pi'}(r,\bar{\boldsymbol{X}}_{r-}^{\pi'})\,\mathrm{d}r\}$. Let $\widetilde{\mathcal{N}}^{\pi'}$ denote the compensated action-marked random measure under π'. The payoff representation of J^π in (19), the p-growth bounds on the stopping payoff and policy entropy, and the moment estimate for the relaxed state process under π give $|J^\pi(s,\boldsymbol{y})| \leq C_\pi(1+|\boldsymbol{y}|^p)$. For $v\in(t,T)$, apply the jump-diffusion Itô formula to $J^\pi(s,\bar{\boldsymbol{X}}_s^{\pi'})$ and the product rule to $\Lambda_s J^\pi(s,\bar{\boldsymbol{X}}_s^{\pi'})$. This gives

$$\begin{aligned}
&\Lambda_v J^\pi(v,\bar{\boldsymbol{X}}_v^{\pi'}) - J^\pi(t,\boldsymbol{x}) + \int_t^v \Lambda_s\left[K\bar{\beta}^{\pi'} g(s,\bar{\boldsymbol{X}}_{s-}^{\pi'}) - \lambda\int_U \pi'(\boldsymbol{u}\mid s,\bar{\boldsymbol{X}}_{s-}^{\pi'})\log\pi'(\boldsymbol{u}\mid s,\bar{\boldsymbol{X}}_{s-}^{\pi'})\,\mathrm{d}\boldsymbol{u}\right]\mathrm{d}s\\
&= \int_t^v \Lambda_s \int_U \Big[J_s^\pi(s,\bar{\boldsymbol{X}}_{s-}^{\pi'}) - \rho J^\pi(s,\bar{\boldsymbol{X}}_{s-}^{\pi'}) + H(s,\bar{\boldsymbol{X}}_{s-}^{\pi'},\boldsymbol{u};J^\pi) - \lambda\log\pi'(\boldsymbol{u}\mid s,\bar{\boldsymbol{X}}_{s-}^{\pi'})\Big]\pi'(\boldsymbol{u}\mid s,\bar{\boldsymbol{X}}_{s-}^{\pi'})\,\mathrm{d}\boldsymbol{u}\,\mathrm{d}s\\
&\quad + \int_t^v \Lambda_s J_{\boldsymbol{x}}^\pi(s,\bar{\boldsymbol{X}}_{s-}^{\pi'})\bar{\boldsymbol{\sigma}}^{\pi'}(s,\bar{\boldsymbol{X}}_{s-}^{\pi'})\,\mathrm{d}\bar{\boldsymbol{W}}_s\\
&\quad + \int_t^v\int_U\int_{\mathbb{R}} \Lambda_s\left[J^\pi\Big(s,\bar{\boldsymbol{X}}_{s-}^{\pi'} + \boldsymbol{\gamma}(s,\bar{\boldsymbol{X}}_{s-}^{\pi'},\boldsymbol{u}^\alpha,z)\Big) - J^\pi(s,\bar{\boldsymbol{X}}_{s-}^{\pi'})\right]\widetilde{\mathcal{N}}^{\pi'}(\mathrm{d}s,\mathrm{d}z,\mathrm{d}\boldsymbol{u}),
\end{aligned} \tag{A4}$$

where, by (26), the expression in square brackets in the first integral on the right is $q(s,\bar{\boldsymbol{X}}_{s-}^{\pi'},\boldsymbol{u};\pi) - \lambda\log\pi'(\boldsymbol{u}\mid s,\bar{\boldsymbol{X}}_{s-}^{\pi'})$. Under the mixed-action integration convention used here, the variational argument for Gibbs maximization is the same as in Jia and Zhou (2023, Lemma 13). Since $q(s,\boldsymbol{y},\boldsymbol{u};\pi)$ and $H(s,\boldsymbol{y},\boldsymbol{u};J^\pi)$ differ only by the action-independent term $J_s^\pi(s,\boldsymbol{y}) - \rho J^\pi(s,\boldsymbol{y})$, π' also maximizes the entropy-regularized objective with q as its action function. Together with the policy evaluation equation (23), this implies, for every $(s,\boldsymbol{y})\in[0,T)\times\mathbb{R}^n$,

$$\begin{aligned}
&\int_U \big[q(s,\boldsymbol{y},\boldsymbol{u};\pi) - \lambda\log\pi'(\boldsymbol{u}\mid s,\boldsymbol{y})\big]\pi'(\boldsymbol{u}\mid s,\boldsymbol{y})\,\mathrm{d}\boldsymbol{u}\\
&\qquad \geq \int_U \big[q(s,\boldsymbol{y},\boldsymbol{u};\pi) - \lambda\log\pi(\boldsymbol{u}\mid s,\boldsymbol{y})\big]\pi(\boldsymbol{u}\mid s,\boldsymbol{y})\,\mathrm{d}\boldsymbol{u} = 0,
\end{aligned} \tag{A5}$$

so the finite-variation term on the right of (A4) is nonnegative.

Choose n sufficiently large and set $\nu_n := \inf\{s \geq t : |\bar{\boldsymbol{X}}_s^{\pi'}| \geq n\}$. For $0 < \varepsilon < T - t$, let $\theta_{n,\varepsilon} := (T - \varepsilon) \wedge \nu_n$. Equation (A4) and the nonnegativity above remain valid when v is replaced by $\theta_{n,\varepsilon}$. Taking expectations and using (A5) gives

$$J^{\pi}(t, \boldsymbol{x}) \leq \mathbb{E}_{t,\boldsymbol{x}}^{\pi'}\left[\Lambda_{\theta_{n,\varepsilon}} J^{\pi}\left(\theta_{n,\varepsilon}, \bar{\boldsymbol{X}}_{\theta_{n,\varepsilon}}^{\pi'}\right) + \int_t^{\theta_{n,\varepsilon}} \Lambda_s \big[K\bar{\beta}^{\pi'} g + \lambda \mathcal{H}(\pi'(\cdot \mid s, \cdot))\big](s, \bar{\boldsymbol{X}}_{s-}^{\pi'})\, \mathrm{d}s\right]. \quad \text{(A6)}$$

Since $0 \leq \bar{\beta}^{\pi'} \leq 1$, Λ is bounded on $[t, T]$, and J^{π}, g and the policy entropy satisfy p-growth bounds, there is a constant $C > 0$, independent of n and ε, such that

$$\left|\Lambda_{\theta_{n,\varepsilon}} J^{\pi}\left(\theta_{n,\varepsilon}, \bar{\boldsymbol{X}}_{\theta_{n,\varepsilon}}^{\pi'}\right)\right| + \int_t^{\theta_{n,\varepsilon}} \Lambda_s \left|K\bar{\beta}^{\pi'} g + \lambda \mathcal{H}(\pi'(\cdot \mid s, \cdot))\right| (s, \bar{\boldsymbol{X}}_{s-}^{\pi'})\, \mathrm{d}s \leq C\left(1 + \sup_{r \in [t,T]} |\bar{\boldsymbol{X}}_r^{\pi'}|^p\right),$$

where the right-hand side is integrable by the moment estimate for the relaxed state process under π'. Applying dominated convergence to (A6), first letting $\varepsilon \downarrow 0$ and then $n \to \infty$, and using the terminal condition $J^{\pi}(T, \boldsymbol{y}) = g(T, \boldsymbol{y})$ yields

$$J^{\pi}(t, \boldsymbol{x}) \leq \mathbb{E}_{t,\boldsymbol{x}}^{\pi'}\left[\Lambda_T g(T, \bar{\boldsymbol{X}}_T^{\pi'}) + \int_t^T \Lambda_s \big[K\bar{\beta}^{\pi'} g + \lambda \mathcal{H}(\pi'(\cdot \mid s, \cdot))\big](s, \bar{\boldsymbol{X}}_s^{\pi'})\, \mathrm{d}s\right] = J^{\pi'}(t, \boldsymbol{x}).$$

At $t = T$, both values equal $g(T, \boldsymbol{x})$. Thus $\pi' = \mathcal{I}(\pi)$ improves upon π.

Furthermore, if $\mathcal{I}(\pi^*) = \pi^*$, write $J^* := J^{\pi^*}$. Gibbs maximization yields, for every $(s, \boldsymbol{y}) \in [0, T) \times \mathbb{R}^n$,

$$\int_U \big[H(s, \boldsymbol{y}, \boldsymbol{u}; J^*) - \lambda \log \pi^*(\boldsymbol{u} \mid s, \boldsymbol{y})\big]\pi^*(\boldsymbol{u} \mid s, \boldsymbol{y})\, \mathrm{d}\boldsymbol{u} = \sup_{\eta \in \mathcal{P}(U)} \int_U \big[H(s, \boldsymbol{y}, \boldsymbol{u}; J^*) - \lambda \log \eta(\boldsymbol{u})\big]\eta(\boldsymbol{u})\, \mathrm{d}\boldsymbol{u}.$$

Substituting this identity into the policy evaluation equation for π^* shows that J^* satisfies the exploratory HJB equation (20) and the terminal condition $J^*(T, \boldsymbol{y}) = g(T, \boldsymbol{y})$. Corollary 3.1 then yields $J^* = v^{\Pi}$, and the fixed-point policy π^* is optimal. □

A.4. *Proof of Theorem 4.1*

Proof. Let η be an arbitrary behavior policy, with $\eta = \pi$ corresponding to the on-policy case. Write $\widetilde{N}(\mathrm{d}r, \mathrm{d}z) := N(\mathrm{d}r, \mathrm{d}z) - m(\mathrm{d}z)\, \mathrm{d}r$. Apply the jump-diffusion Itô formula to $J^{\pi}(s, \boldsymbol{X}_s^{\eta})$ and the product rule to $\mathrm{e}^{-\rho(s-t)} R_s^{\eta} J^{\pi}(s, \boldsymbol{X}_s^{\eta})$. Since $\mathrm{d}R_s^{\eta} = -K R_s^{\eta} u_s^{\beta,\eta}\, \mathrm{d}s$, for $t \leq s < T$ we obtain

$$\begin{aligned}
&\mathrm{e}^{-\rho(s-t)} R_s^{\eta} J^{\pi}(s, \boldsymbol{X}_s^{\eta}) - J^{\pi}(t, \boldsymbol{x}) + \int_t^s \mathrm{e}^{-\rho(r-t)} R_r^{\eta}\big[K u_r^{\beta,\eta} g(r, \boldsymbol{X}_{r-}^{\eta}) - \widehat{q}(r, \boldsymbol{X}_{r-}^{\eta}, \boldsymbol{u}_r^{\eta})\big]\, \mathrm{d}r \\
&= \int_t^s \mathrm{e}^{-\rho(r-t)} R_r^{\eta}\Big\{\big[J_r^{\pi} + \mathcal{L}^{\boldsymbol{u}_r^{\alpha,\eta}} J^{\pi} + \mathcal{J}^{\boldsymbol{u}_r^{\alpha,\eta}} J^{\pi} - \rho J^{\pi} + K u_r^{\beta,\eta}(g - J^{\pi})\big](r, \boldsymbol{X}_{r-}^{\eta}) - \widehat{q}(r, \boldsymbol{X}_{r-}^{\eta}, \boldsymbol{u}_r^{\eta})\Big\}\, \mathrm{d}r \\
&\quad + \int_t^s \mathrm{e}^{-\rho(r-t)} R_r^{\eta} J_{\boldsymbol{x}}^{\pi}(r, \boldsymbol{X}_{r-}^{\eta}) \boldsymbol{\sigma}(r, \boldsymbol{X}_{r-}^{\eta}, \boldsymbol{u}_r^{\alpha,\eta})\, \mathrm{d}\boldsymbol{W}_r \\
&\quad + \int_t^s \int_{\mathbb{R}} \mathrm{e}^{-\rho(r-t)} R_r^{\eta}\big[J^{\pi}\big(r, \boldsymbol{X}_{r-}^{\eta} + \boldsymbol{\gamma}(r, \boldsymbol{X}_{r-}^{\eta}, \boldsymbol{u}_r^{\alpha,\eta}, z)\big) - J^{\pi}(r, \boldsymbol{X}_{r-}^{\eta})\big]\widetilde{N}(\mathrm{d}r, \mathrm{d}z).
\end{aligned} \quad \text{(A7)}$$

By (26), the dr integral on the right equals

$$\int_t^s \mathrm{e}^{-\rho(r-t)} R_r^\eta \big[q(r, \boldsymbol{X}_{r-}^\eta, \boldsymbol{u}_r^\eta; \pi) - \widehat{q}(r, \boldsymbol{X}_{r-}^\eta, \boldsymbol{u}_r^\eta)\big]\, \mathrm{d}r.$$

If $\widehat{q} = q(\cdot,\cdot,\cdot;\pi)$, this drift vanishes. Under the stated integrability of the stochastic integrals, the resulting process is an $(\mathbb{F}^{\mathbb{Q}}, \mathbb{Q})$-martingale. Taking $\eta = \pi$ and $\eta = \pi'$, respectively, proves the forward direction of *(i)* and assertion *(ii)*.

Conversely, suppose that (27) is a martingale for every initial point and every time grid. By (A7), the process

$$\int_t^s \mathrm{e}^{-\rho(r-t)} R_r^\pi \big[q(r, \boldsymbol{X}_{r-}^\pi, \boldsymbol{u}_r^\pi; \pi) - \widehat{q}(r, \boldsymbol{X}_{r-}^\pi, \boldsymbol{u}_r^\pi)\big]\, \mathrm{d}r$$

is also a local martingale. A continuous local martingale of finite variation is constant. Since this process starts at zero,

$$\int_t^s \mathrm{e}^{-\rho(r-t)} R_r^\pi \big[q(r, \boldsymbol{X}_{r-}^\pi, \boldsymbol{u}_r^\pi; \pi) - \widehat{q}(r, \boldsymbol{X}_{r-}^\pi, \boldsymbol{u}_r^\pi)\big]\, \mathrm{d}r = 0, \quad s \in [t,T], \quad \mathbb{Q}\text{-a.s.} \tag{A8}$$

Write $h(s,\boldsymbol{x},\boldsymbol{u}) := q(s,\boldsymbol{x},\boldsymbol{u};\pi) - \widehat{q}(s,\boldsymbol{x},\boldsymbol{u})$. If h is not identically zero, we may assume that there are $(t^*,\boldsymbol{x}^*,\boldsymbol{u}^*) \in [0,T) \times \mathbb{R}^n \times U$ and $\varepsilon > 0$ such that $h(t^*,\boldsymbol{x}^*,\boldsymbol{u}^*) > \varepsilon$; the negative case follows by applying the same argument to $-h$. By continuity, there is $\delta > 0$ such that $h(r,\boldsymbol{y},\boldsymbol{u}) > \varepsilon/2$ whenever $|r-t^*| \vee |\boldsymbol{y}-\boldsymbol{x}^*| \vee |\boldsymbol{u}-\boldsymbol{u}^*| < \delta$.

Consider the sampled state process starting from $\boldsymbol{X}_{t^*}^\pi = \boldsymbol{x}^*$, and choose a time grid satisfying $t^* = t_0 < t^* + \delta < t_1$. Define $\tau := \inf\{r \geq t^* : |\boldsymbol{X}_r^\pi - \boldsymbol{x}^*| \geq \delta\} \wedge (t^* + \delta)$. The right-continuity of the jump-diffusion paths implies $\tau > t^*$ almost surely. By (A8), there is a $\mathbb{Q}$-null set Ω_0 outside which the corresponding integral from t^* to s vanishes for every $s \in [t^*, T]$. Since $\mathrm{e}^{-\rho(r-t^*)} R_r^\pi > 0$, the Lebesgue differentiation theorem yields, for each $\omega \notin \Omega_0$,

$$h(r, \boldsymbol{X}_{r-}^\pi(\omega), \boldsymbol{u}_r^\pi(\omega)) = 0 \quad \text{for almost every } r \in [t^*, \tau(\omega)].$$

On the other hand, on the first grid interval, $\boldsymbol{u}_r^\pi = \boldsymbol{u}_{t^*}^\pi = \Phi(t^*,\boldsymbol{x}^*,\xi_0)$. Let $B_\delta^U(\boldsymbol{u}^*) := \{\boldsymbol{u} \in U : |\boldsymbol{u}-\boldsymbol{u}^*| < \delta\}$. The full-support property of π gives $\mathbb{Q}(\boldsymbol{u}_{t^*}^\pi \in B_\delta^U(\boldsymbol{u}^*)) = \int_{B_\delta^U(\boldsymbol{u}^*)} \pi(\boldsymbol{u} \mid t^*, \boldsymbol{x}^*)\, \mathrm{d}\boldsymbol{u} > 0$. We can therefore select a path in this positive-probability event outside Ω_0 for which $\tau(\omega) > t^*$. By the definition of τ, for $r \in [t^*, \tau(\omega)]$, we have $|r - t^*| < \delta$ and $|\boldsymbol{X}_r^\pi(\omega) - \boldsymbol{x}^*| < \delta$. A càdlàg jump-diffusion path has at most countably many jump times, so $\boldsymbol{X}_{r-}^\pi = \boldsymbol{X}_r^\pi$ for almost every r. Consequently, for almost every $r \in [t^*, \tau(\omega)]$,

$$h(r, \boldsymbol{X}_{r-}^\pi(\omega), \boldsymbol{u}_r^\pi(\omega)) = h(r, \boldsymbol{X}_r^\pi(\omega), \Phi(t^*, \boldsymbol{x}^*, \xi_0(\omega))) > \varepsilon/2 > 0,$$

which contradicts its almost-everywhere vanishing on the same interval. Thus $h \equiv 0$, that is, $\widehat{q}(s,\boldsymbol{x},\boldsymbol{u}) = q(s,\boldsymbol{x},\boldsymbol{u};\pi)$ for every $(s,\boldsymbol{x},\boldsymbol{u}) \in [0,T] \times \mathbb{R}^n \times U$.

If (28) satisfies the condition in *(iii)*, apply the same decomposition with $\eta = \pi'$. Since π' also has full support, the same contradiction argument gives $\widehat{q} = q(\cdot,\cdot,\cdot;\pi)$. Finally, (29) follows from (26) and the policy evaluation equation (23). □

A.5. *Proof of Theorem 4.2*

Proof. We first prove *(i)*. If $\widehat{J} = J^\pi$ and $\widehat{q} = q(\cdot,\cdot,\cdot;\pi)$, the stated martingale property follows directly from Theorem 4.1*(i)*. Conversely, suppose that the process in (31) is a martingale for

every initial point and time grid. Define

$$\widehat{r}(s,\boldsymbol{y},\boldsymbol{u}) := \widehat{J}_s(s,\boldsymbol{y}) + \mathcal{L}^{\boldsymbol{u}^\alpha}\widehat{J}(s,\boldsymbol{y}) + \mathcal{J}^{\boldsymbol{u}^\alpha}\widehat{J}(s,\boldsymbol{y}) - (\rho + Ku^\beta)\widehat{J}(s,\boldsymbol{y}),$$

and consider

$$M_s^{\widehat{J}} := \mathrm{e}^{-\rho(s-t)}R_s^\pi \widehat{J}(s,\boldsymbol{X}_s^\pi) - \int_t^s \mathrm{e}^{-\rho(r-t)}R_r^\pi \widehat{r}(r,\boldsymbol{X}_{r-}^\pi,\boldsymbol{u}_r^\pi)\,\mathrm{d}r.$$

The jump-diffusion Itô and product formulas applied to $\mathrm{e}^{-\rho(s-t)}R_s^\pi \widehat{J}(s,\boldsymbol{X}_s^\pi)$ give

$$\begin{aligned} M_s^{\widehat{J}} = \widehat{J}(t,\boldsymbol{x}) &+ \int_t^s \mathrm{e}^{-\rho(r-t)}R_r^\pi \widehat{J}_{\boldsymbol{x}}(r,\boldsymbol{X}_{r-}^\pi)\boldsymbol{\sigma}(r,\boldsymbol{X}_{r-}^\pi,\boldsymbol{u}_r^{\alpha,\pi})\,\mathrm{d}\boldsymbol{W}_r \\ &+ \int_t^s\int_{\mathbb{R}} \mathrm{e}^{-\rho(r-t)}R_r^\pi \big[\widehat{J}\big(r,\boldsymbol{X}_{r-}^\pi + \boldsymbol{\gamma}(r,\boldsymbol{X}_{r-}^\pi,\boldsymbol{u}_r^{\alpha,\pi},z)\big) - \widehat{J}(r,\boldsymbol{X}_{r-}^\pi)\big]\widetilde{N}(\mathrm{d}r,\mathrm{d}z), \end{aligned}$$

so $M^{\widehat{J}}$ is a local martingale. The difference $\widehat{M}_s - M_s^{\widehat{J}}$ is

$$\int_t^s \mathrm{e}^{-\rho(r-t)}R_r^\pi \big[Ku_r^{\beta,\pi}g(r,\boldsymbol{X}_{r-}^\pi) - \widehat{q}(r,\boldsymbol{X}_{r-}^\pi,\boldsymbol{u}_r^\pi) + \widehat{r}(r,\boldsymbol{X}_{r-}^\pi,\boldsymbol{u}_r^\pi)\big]\,\mathrm{d}r.$$

This difference is a continuous finite-variation local martingale starting from zero and is therefore identically zero. Applying the reverse-direction argument in Theorem 4.1*(i)* gives, for every $(s,\boldsymbol{y},\boldsymbol{u}) \in [0,T]\times\mathbb{R}^n\times U$,

$$\begin{aligned} \widehat{q}(s,\boldsymbol{y},\boldsymbol{u}) &= Ku^\beta g(s,\boldsymbol{y}) + \widehat{r}(s,\boldsymbol{y},\boldsymbol{u}) \\ &= \widehat{J}_s(s,\boldsymbol{y}) + \mathcal{L}^{\boldsymbol{u}^\alpha}\widehat{J}(s,\boldsymbol{y}) + \mathcal{J}^{\boldsymbol{u}^\alpha}\widehat{J}(s,\boldsymbol{y}) - \rho\widehat{J}(s,\boldsymbol{y}) + Ku^\beta\big(g(s,\boldsymbol{y}) - \widehat{J}(s,\boldsymbol{y})\big). \end{aligned} \tag{A9}$$

Substituting (A9) into the entropy-consistency condition in (30) shows that $\widehat{J}$ satisfies

$$\begin{cases} \widehat{J}_s(s,\boldsymbol{y}) - \rho\widehat{J}(s,\boldsymbol{y}) + \displaystyle\int_U \big[H(s,\boldsymbol{y},\boldsymbol{u};\widehat{J}) - \lambda\log\pi(\boldsymbol{u}\mid s,\boldsymbol{y})\big]\pi(\boldsymbol{u}\mid s,\boldsymbol{y})\,\mathrm{d}\boldsymbol{u} = 0, \\ \widehat{J}(T,\boldsymbol{y}) = g(T,\boldsymbol{y}), \end{cases}$$

which is the policy evaluation equation (23) for π. Its verification-based uniqueness in the stated p-growth class gives $\widehat{J} = J^\pi$. The reverse conclusion of Theorem 4.1*(i)* then yields $\widehat{q} = q(\cdot,\cdot,\cdot;\pi)$.

Assertion *(ii)* follows directly from $\widehat{J} = J^\pi$, $\widehat{q} = q(\cdot,\cdot,\cdot;\pi)$ and Theorem 4.1*(ii)*. For *(iii)*, repeating the reverse-direction argument of *(i)* along the sampled process under the behavior policy π' yields (A9). Substituting that identity into (30) for the target policy π yields the same policy evaluation equation, and hence $\widehat{J} = J^\pi$. Theorem 4.1 *(iii)* then gives $\widehat{q} = q(\cdot,\cdot,\cdot;\pi)$. Finally, if π also satisfies the Gibbs formula in the theorem, the identity $\widehat{q} = q(\cdot,\cdot,\cdot;\pi)$ shows that $\mathcal{I}(\pi) = \pi$. By the fixed-point conclusion of Theorem 3.2, π is optimal and $\widehat{J} = J^\pi = v^\Pi$. □

Appendix B: Finite-difference method

This section describes the finite-difference benchmark used in Section 6. As in the finite-horizon stopping problem of Dai *et al.* (2026), we use the penalty method of Forsyth and Vetzal (2002) to handle the obstacle condition. Here, however, the penalty equation (5) also contains a pointwise

maximization over investment controls and a nonlocal jump term. We fix $K > 0$, solve the penalty equation using known market parameters, and use its solution to approximate that of the original variational inequality.

Let $t_i = i\Delta t$, with $\Delta t = T/N_t$, and introduce a grid on the truncated wealth interval $[K_0, x_{\max}]$: $x_j = K_0 + j\Delta x$, where $\Delta x = (x_{\max} - K_0)/N_x$. Write $V_{ij} \approx v^K(t_i, x_j)$ and $g_j = g(x_j)$. The terminal and left boundary values are $V_{N_t,j} = g_j$ and $V_{i0} = 0$. At the right boundary, use

$$V_{i,N_x} = \varphi(t_i, x_{\max}), \quad \varphi(t, x_{\max}) = \max\Big\{g(x_{\max}), \mathrm{e}^{-\rho(T-t)} g\Big(x_{\max}\mathrm{e}^{r(T-t)}\Big)\Big\}. \tag{B1}$$

For $\alpha \in [0,1]$, set $a_j(\alpha) = \frac{1}{2}\alpha^2\sigma^2 x_j^2$ and $c_j(\alpha) = -x_j[r + \alpha(\mu - r - \nu J)]$, where c_j is the first-derivative coefficient after moving the drift term to the left of the penalty equation. We use an implicit time difference when solving backward, a central difference for the second spatial derivative, and an upwind difference for the first spatial derivative:

$$\begin{aligned} \partial_t v^K(t_i, x_j) &= \frac{V_{i+1,j} - V_{ij}}{\Delta t}, \quad D_{xx}V_{ij} = \frac{V_{i,j+1} - 2V_{ij} + V_{i,j-1}}{(\Delta x)^2}, \\ cD_x^{\mathrm{up},c}V_{ij} &= c^+\frac{V_{ij} - V_{i,j-1}}{\Delta x} + c^-\frac{V_{i,j+1} - V_{ij}}{\Delta x}, \end{aligned} \tag{B2}$$

where $c^+ = \max\{c, 0\}$ and $c^- = \min\{c, 0\}$. Post-jump wealth $x_j(1 + \alpha J)$ generally falls between grid points. Let $\mathcal{I}_i^{(k)}(y)$ denote linear interpolation of the kth iterate $\boldsymbol{V}_i^{(k)}$ at y. For $y \le K_0$ or $y \ge x_{\max}$, use 0 or $\varphi(t_i, x_{\max})$, respectively.

Given the next time layer $\boldsymbol{V}_{i+1}$, initialize the current layer with $\boldsymbol{V}_i^{(0)} = \boldsymbol{V}_{i+1}$. At iteration k, choose the control at each node from the grid $0 = \alpha_0 < \cdots < \alpha_M = 1$:

$$\alpha_{ij}^{(k)} \in \arg\max_{\alpha_m}\Big\{a_j(\alpha_m)D_{xx}V_{ij}^{(k)} - c_j(\alpha_m)D_x^{\mathrm{up},c_j(\alpha_m)}V_{ij}^{(k)} + \nu\big[\mathcal{I}_i^{(k)}(x_j(1+\alpha_m J)) - V_{ij}^{(k)}\big]\Big\}. \tag{B3}$$

The control-independent discount and penalty terms do not affect this maximization. Set $I_{ij}^{(k)} = \mathbf{1}_{\{g_j > V_{ij}^{(k)}\}}$, $a_{ij}^{(k)} = a_j(\alpha_{ij}^{(k)})$ and $c_{ij}^{(k)} = c_j(\alpha_{ij}^{(k)})$. To retain a tridiagonal linear system at each iteration, we evaluate post-jump arrival values using $\mathcal{I}_i^{(k)}$ and treat the jump loss at the current node implicitly. Define

$$\ell_{ij}^{(k)} := \frac{a_{ij}^{(k)}}{(\Delta x)^2} + \frac{(c_{ij}^{(k)})^+}{\Delta x}, \quad r_{ij}^{(k)} := \frac{a_{ij}^{(k)}}{(\Delta x)^2} - \frac{(c_{ij}^{(k)})^-}{\Delta x}. \tag{B4}$$

For $j = 1, \ldots, N_x - 1$, the time–diffusion–jump operator with the control fixed at its current iterate acts on $\boldsymbol{V}_i^{(k+1)}$ as

$$\begin{aligned} \mathcal{L}_{ij}^{(k)}\boldsymbol{V}_i^{(k+1)} = \frac{V_{i+1,j} - V_{ij}^{(k+1)}}{\Delta t} &+ \ell_{ij}^{(k)}V_{i,j-1}^{(k+1)} + r_{ij}^{(k)}V_{i,j+1}^{(k+1)} \\ &- \big(\ell_{ij}^{(k)} + r_{ij}^{(k)} + \rho + \nu\big)V_{ij}^{(k+1)} + \nu\mathcal{I}_i^{(k)}\Big(x_j(1+\alpha_{ij}^{(k)}J)\Big), \end{aligned} \tag{B5}$$

and the discrete penalty equation is

$$-\mathcal{L}_{ij}^{(k)}\boldsymbol{V}_i^{(k+1)} = K(g_j - V_{ij}^{(k+1)})^+, \tag{B6}$$

whose penalty term is linearized using $I_{ij}^{(k)}$ as

$$(g_j - V_{ij}^{(k+1)})^+ \approx (g_j - V_{ij}^{(k)})^+ - I_{ij}^{(k)}(V_{ij}^{(k+1)} - V_{ij}^{(k)}) = I_{ij}^{(k)}(g_j - V_{ij}^{(k+1)}). \tag{B7}$$

Substituting (B5) and (B7) into (B6) gives

$$\begin{aligned} &-\ell_{ij}^{(k)} V_{i,j-1}^{(k+1)} + \left[\frac{1}{\Delta t} + \ell_{ij}^{(k)} + r_{ij}^{(k)} + \rho + \nu + K I_{ij}^{(k)}\right] V_{ij}^{(k+1)} - r_{ij}^{(k)} V_{i,j+1}^{(k+1)} \\ &\quad = \frac{V_{i+1,j}}{\Delta t} + K I_{ij}^{(k)} g_j + \nu \mathcal{I}_i^{(k)}\Big(x_j(1+\alpha_{ij}^{(k)} J)\Big). \end{aligned} \tag{B8}$$

At the interior nodes adjacent to the boundaries, $j = 1$ and $j = N_x - 1$, move the terms involving the prescribed boundary values to the right-hand side of (B8). The resulting right-hand side depends only on the solution at the next time layer and quantities available before the current iteration. The interior values $\boldsymbol{V}_i^{(k+1)}$ are therefore obtained by solving a tridiagonal linear system at each iteration. Algorithm 2 summarizes the finite-difference procedure for solving the penalized equation.

Algorithm 2: Finite-difference solution of the penalty equation

Input: $\Delta t, \Delta x, N_t, N_x, x_{\max}, K$, control grid $\{\alpha_m\}_{m=0}^{M}$, and tolerance $\varepsilon > 0$
Set $V_{N_t,j} = g_j$; set the boundary values at each time layer using (B1) and $V_{i0} = 0$;
for $i = N_t - 1, \ldots, 0$ **do**
  Set $V_{ij}^{(0)} = V_{i+1,j}$ for $j = 1, \ldots, N_x - 1$;
  for $k = 0, 1, \ldots$ **do**
    At each interior node, choose $\alpha_{ij}^{(k)}$ using (B3); compute $I_{ij}^{(k)}$ and the post-jump interpolation;
    Solve (B8) for all interior values $V_{ij}^{(k+1)}$;
    if $\|\boldsymbol{V}_i^{(k+1)} - \boldsymbol{V}_i^{(k)}\|_\infty < \varepsilon \max\{1, \|\boldsymbol{V}_i^{(k)}\|_\infty\}$ **then**
      Exit the inner loop;
  Set $V_{ij} = V_{ij}^{(k+1)}$ and store $\alpha_{ij}^{(k)}$ for $j = 1, \ldots, N_x - 1$.